Lab 02: Hands-On Git and Version Control

Name: Fatima Kalsoom

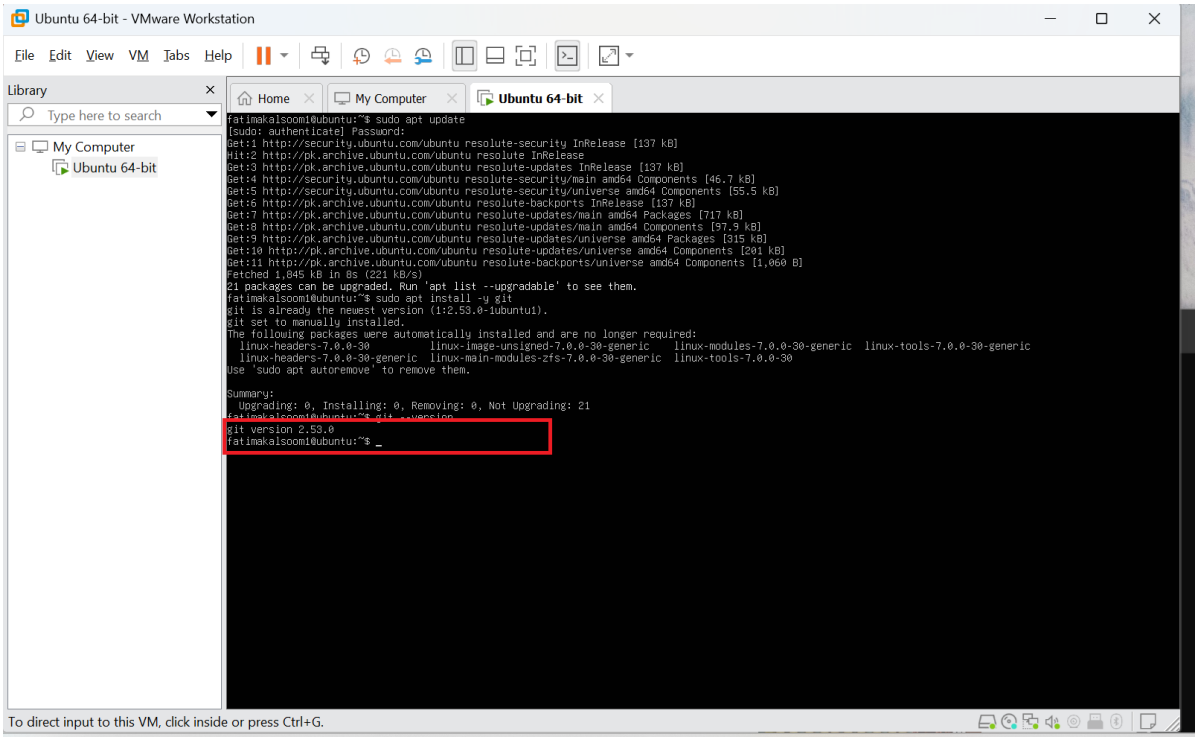
Registration Number: 2023-BSCS-32

Course Code: CS-649

Section: A

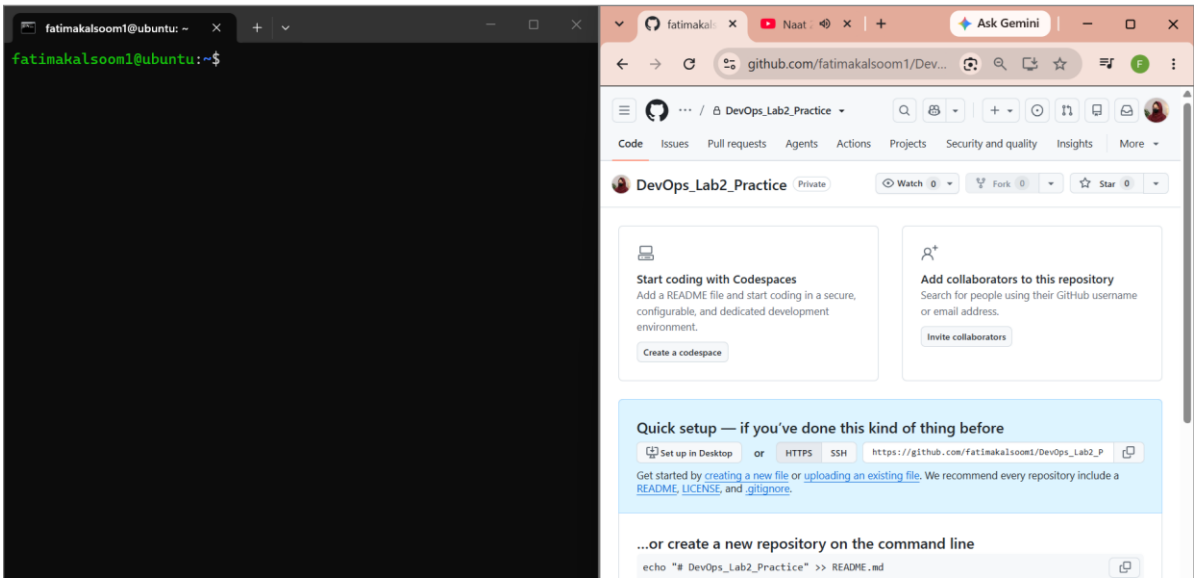
GitHub Username: fatimakalsoom1

Getting Started



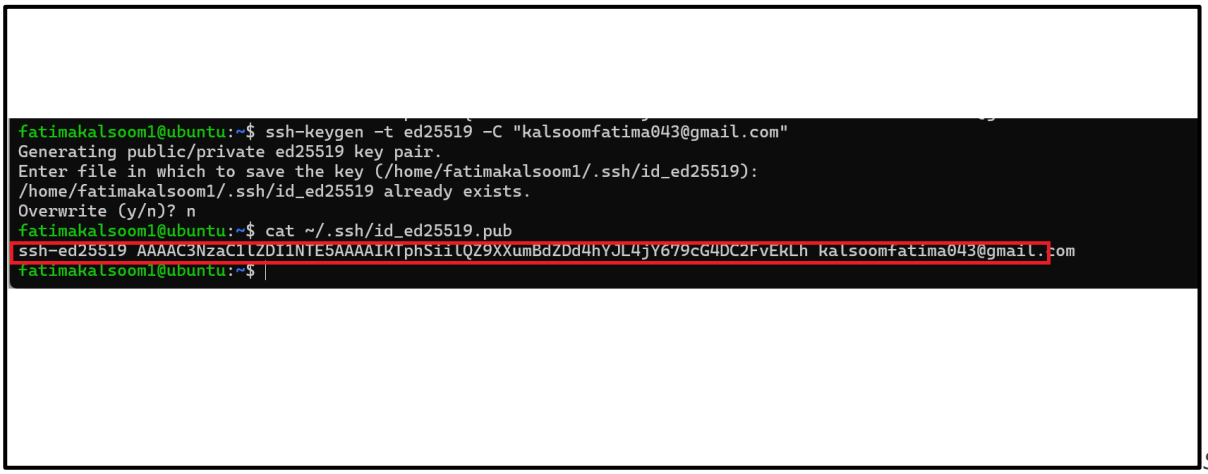
Git installed successfully; version displayed in terminal.

Task 1

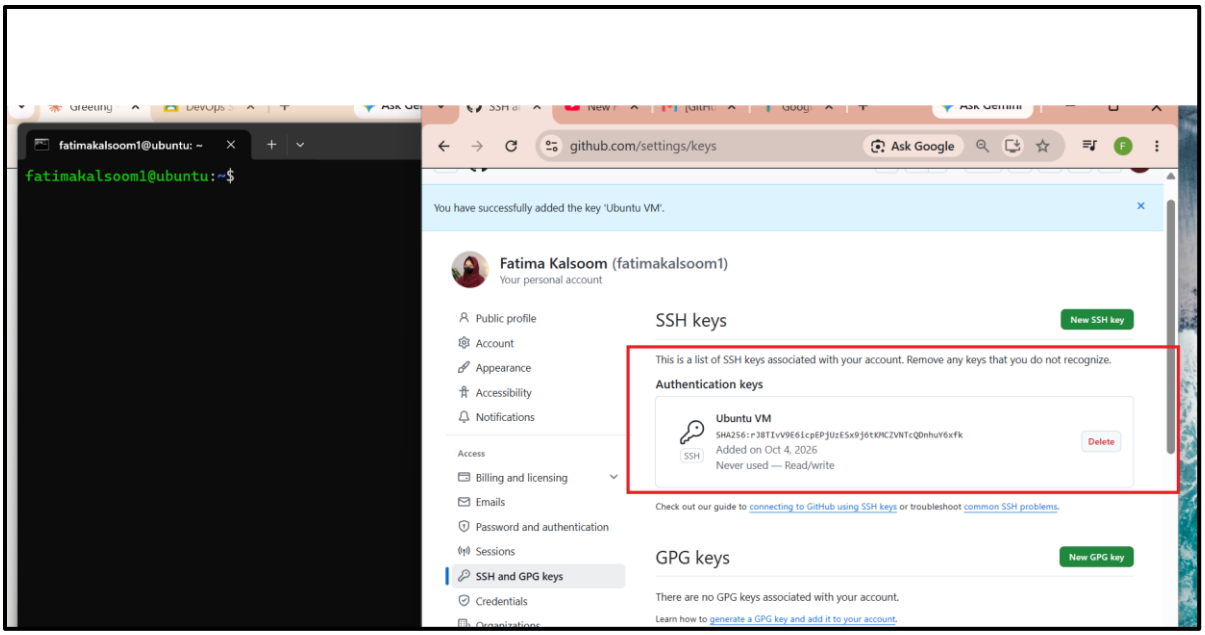


Private repository DevOps_Lab2_Practice created on GitHub.

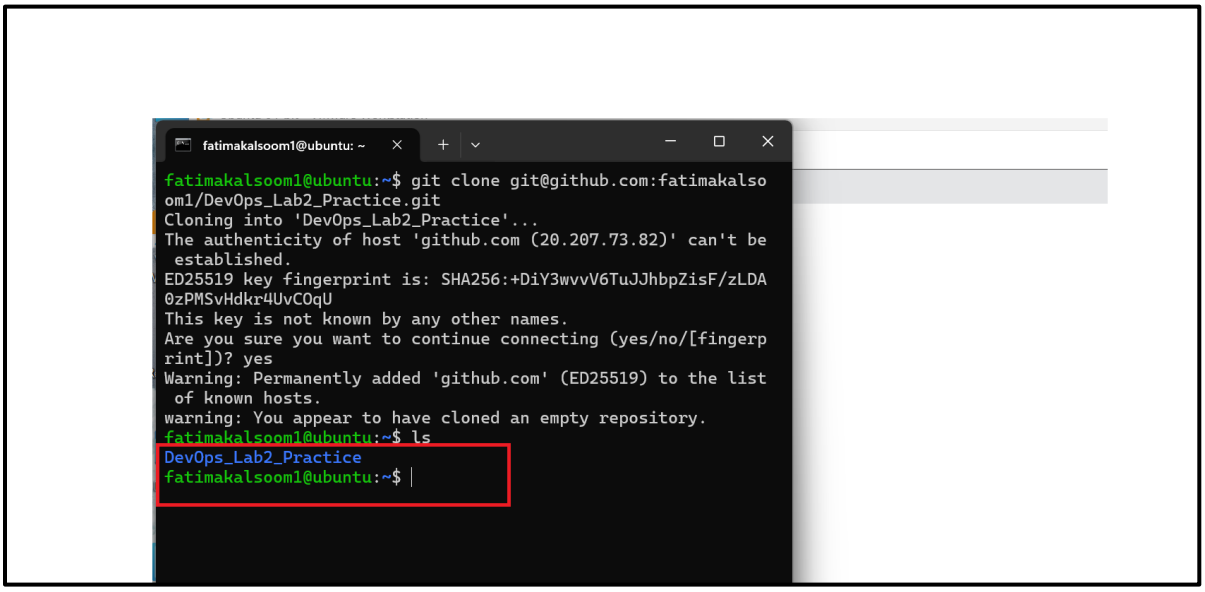
Task 2



SSH key pair generated; public key displayed.

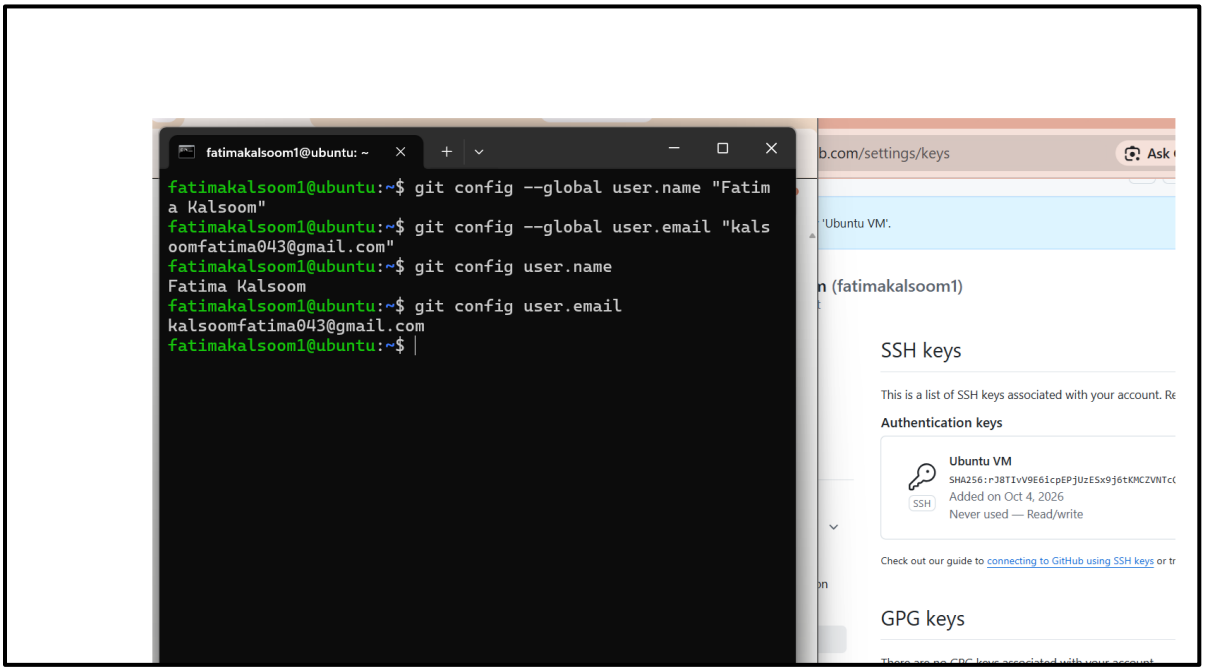


Public SSH key added to GitHub account settings.

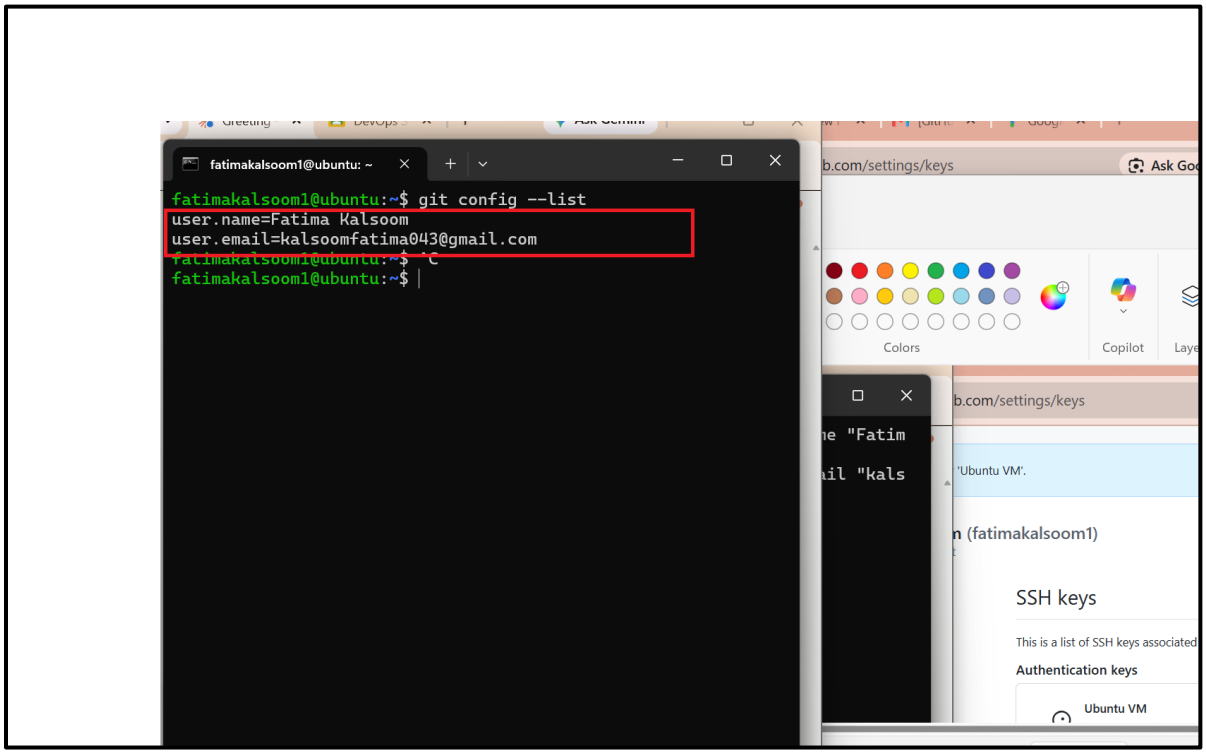


Repository cloned using SSH.

Task 3

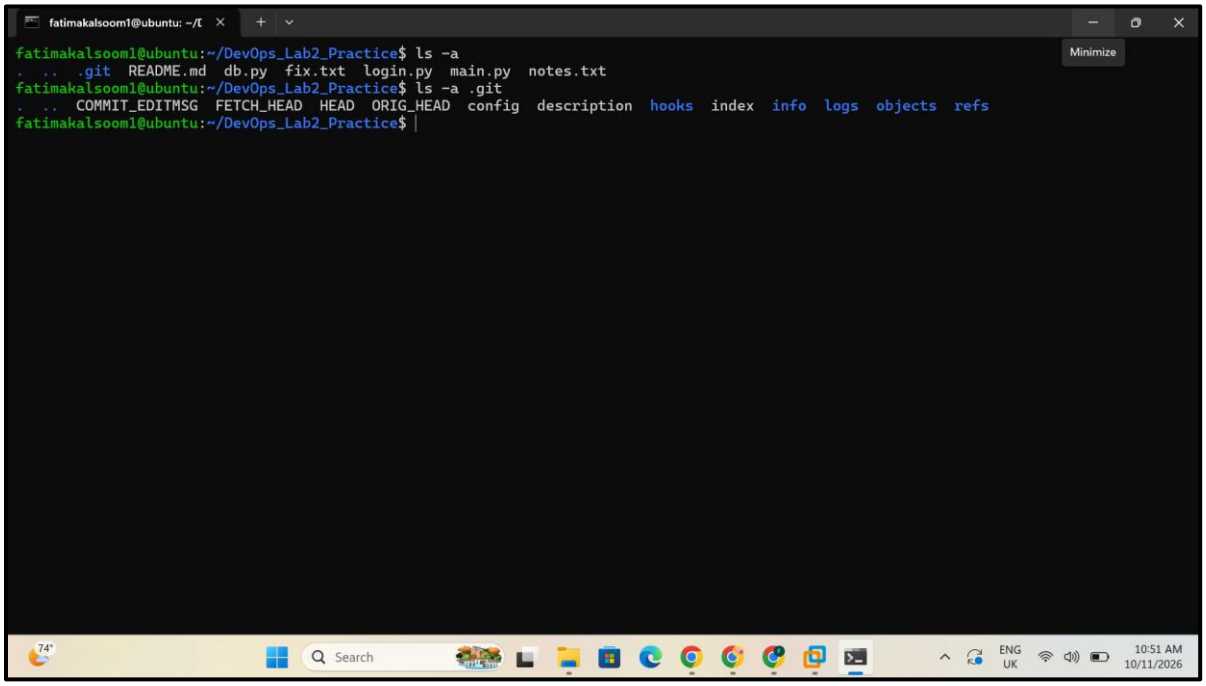


Git username and email configured.



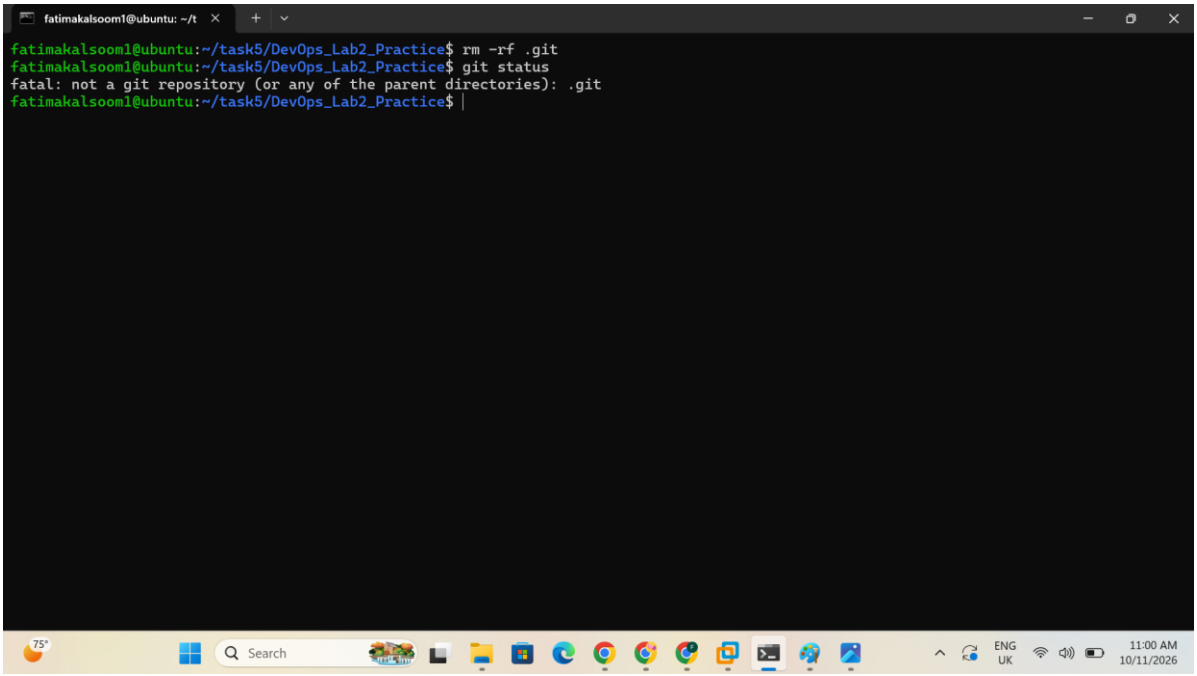
Git configuration verified with git config --list.

Task 4

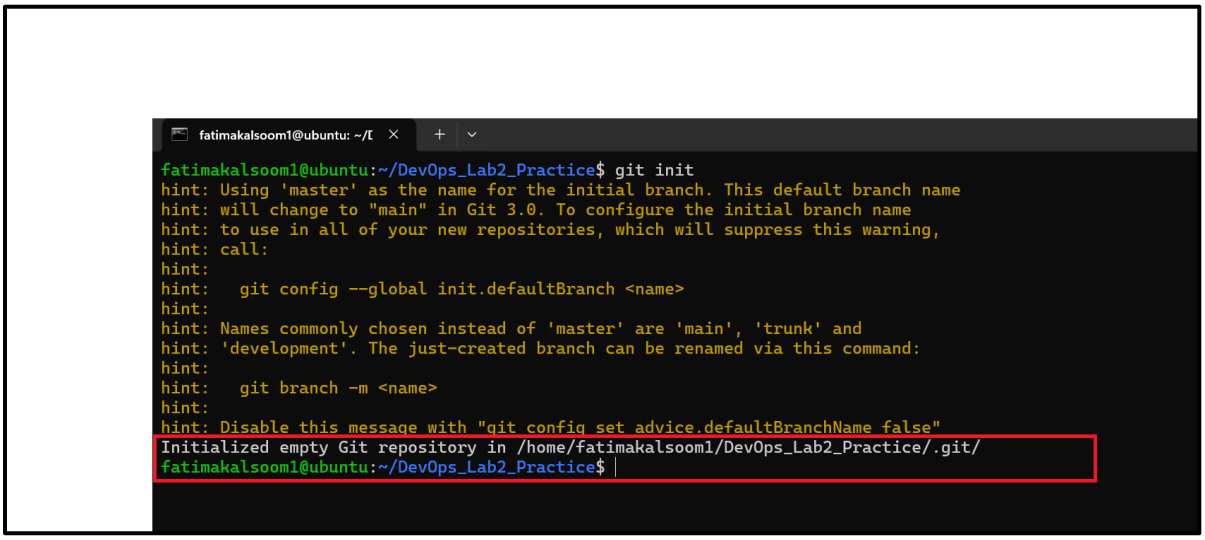


Hidden .git folder and its contents listed.

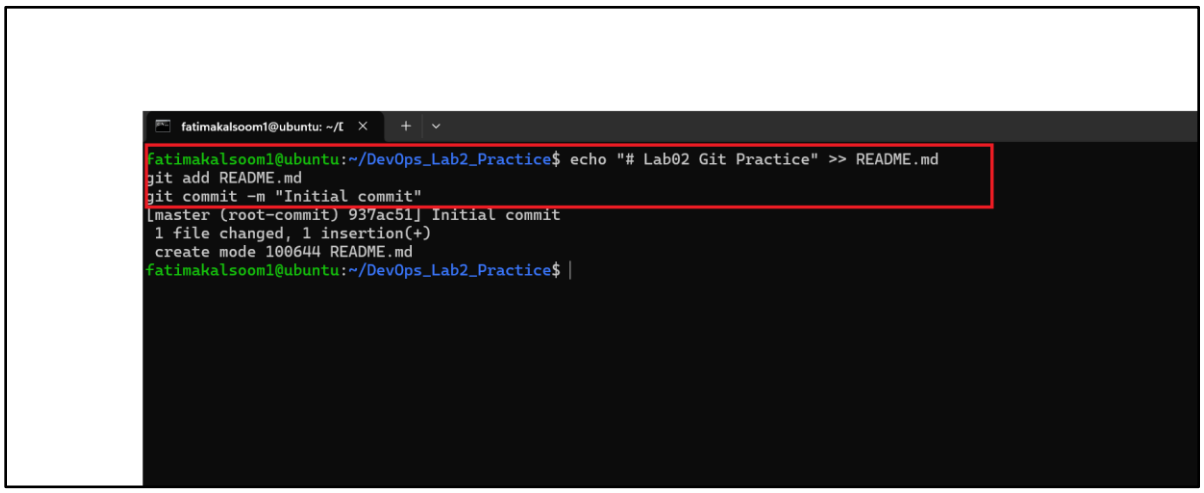
Task 5



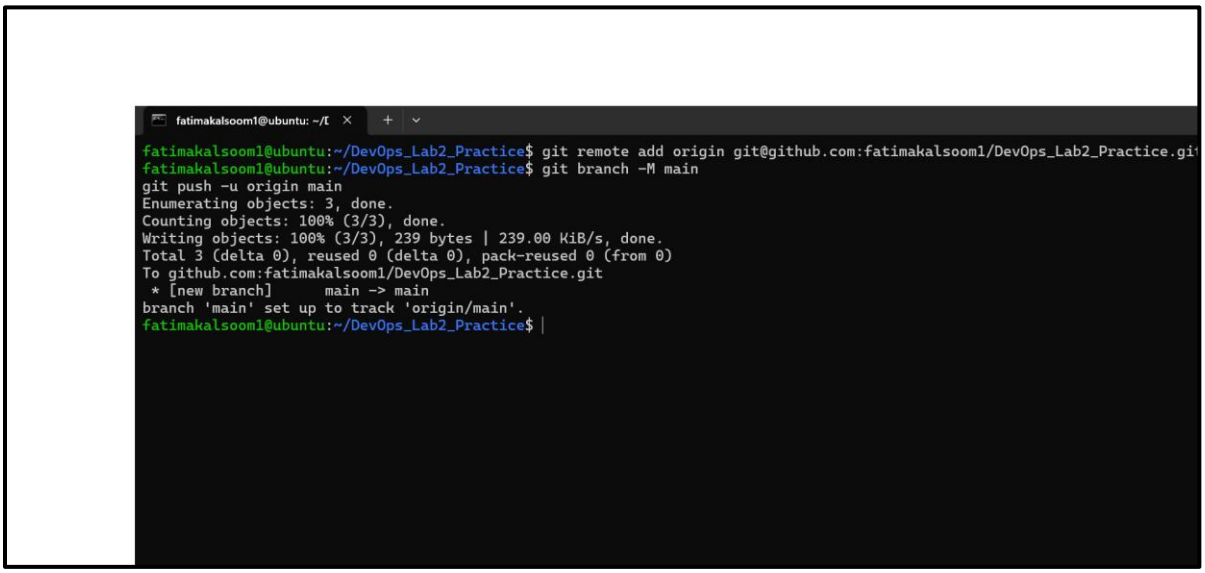
.git folder deleted; git status fails because the folder is no longer a repository.



Empty Git repository re-initialized.

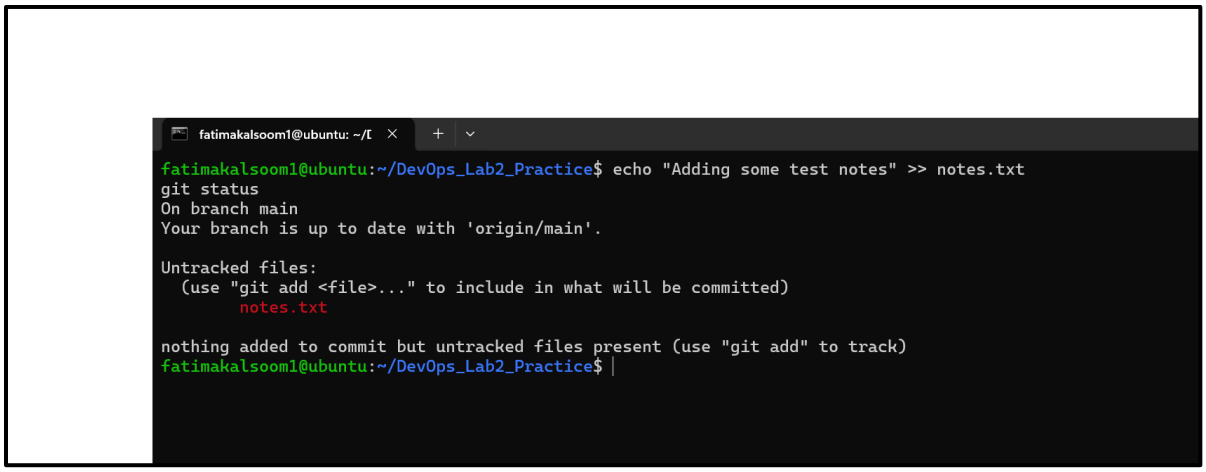


README.md added and initial commit created.

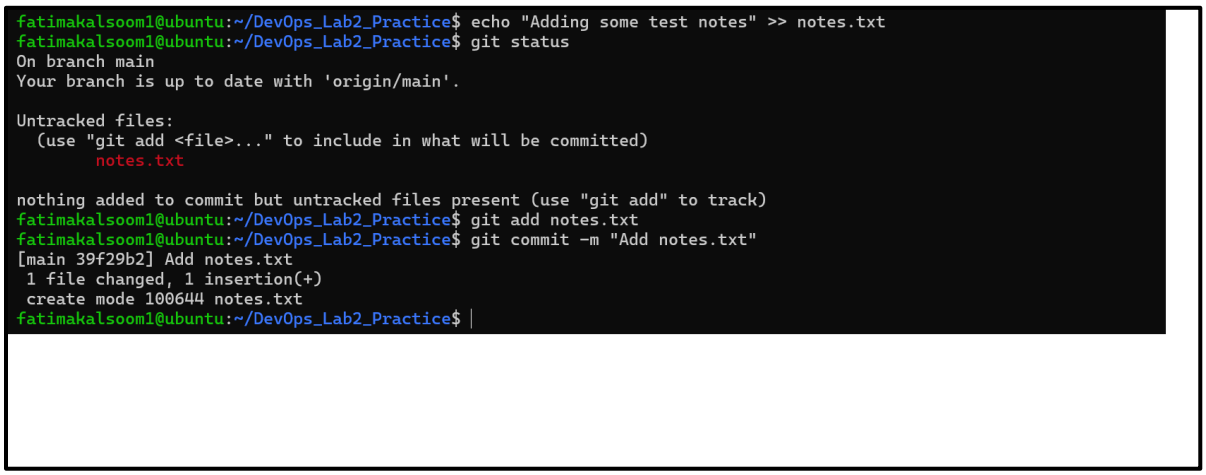


Local repository connected to GitHub and pushed to main.

Task 6

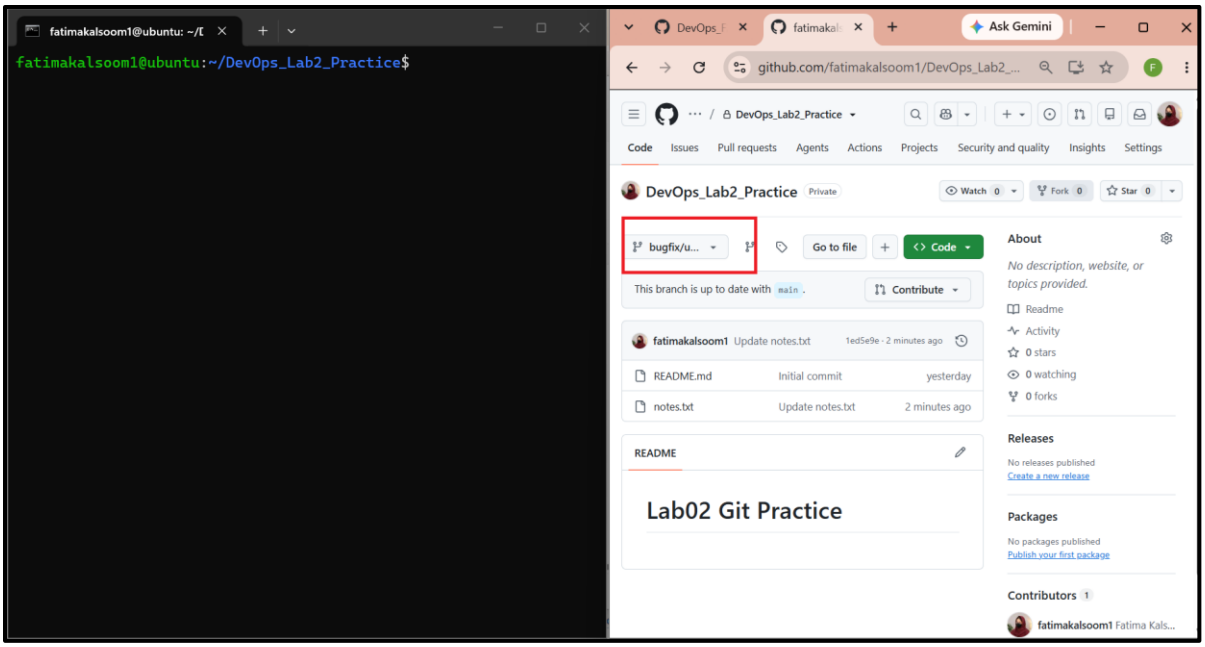


notes.txt shown as untracked in git status.

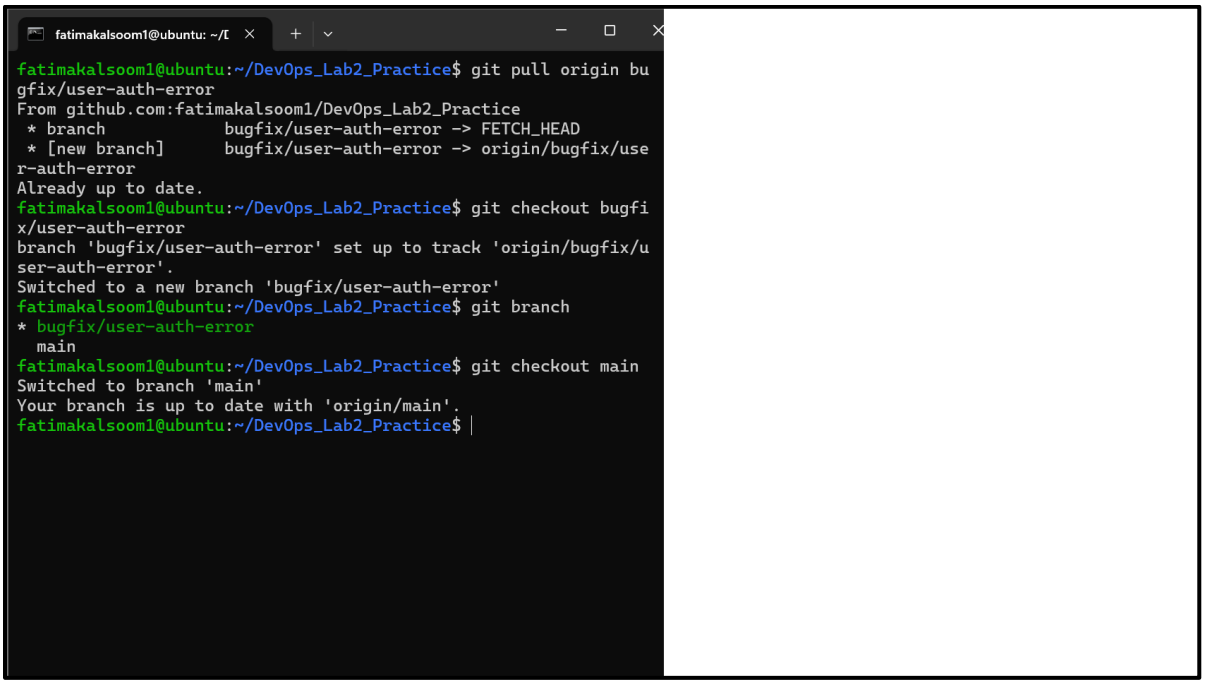


notes.txt staged and committed.

Task 7

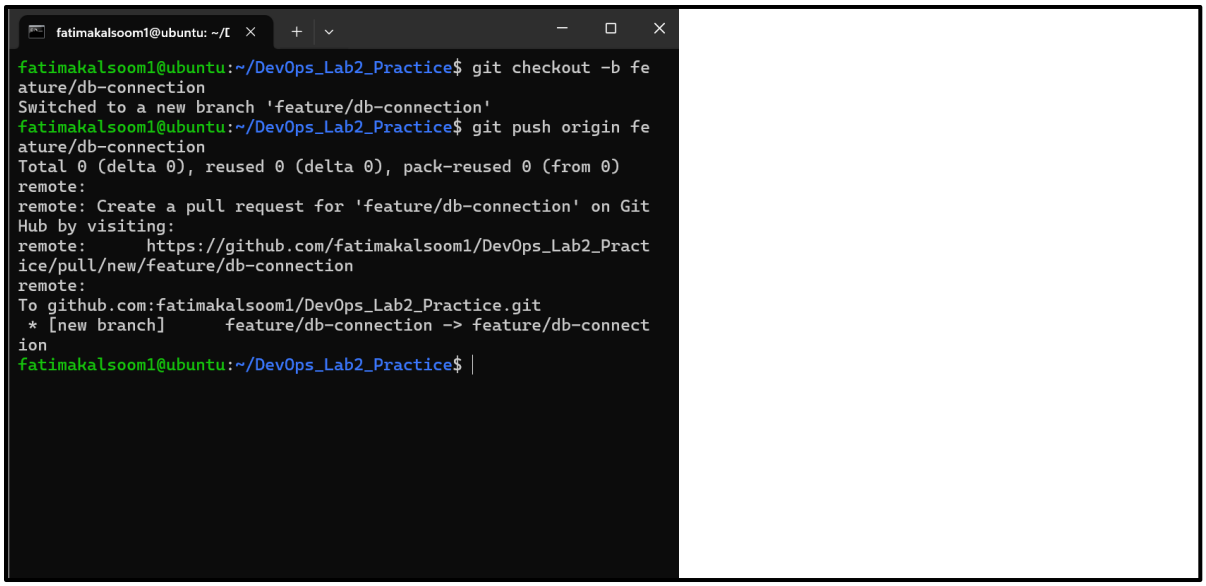


Branch bugfix/user-auth-error created from the GitHub web interface.



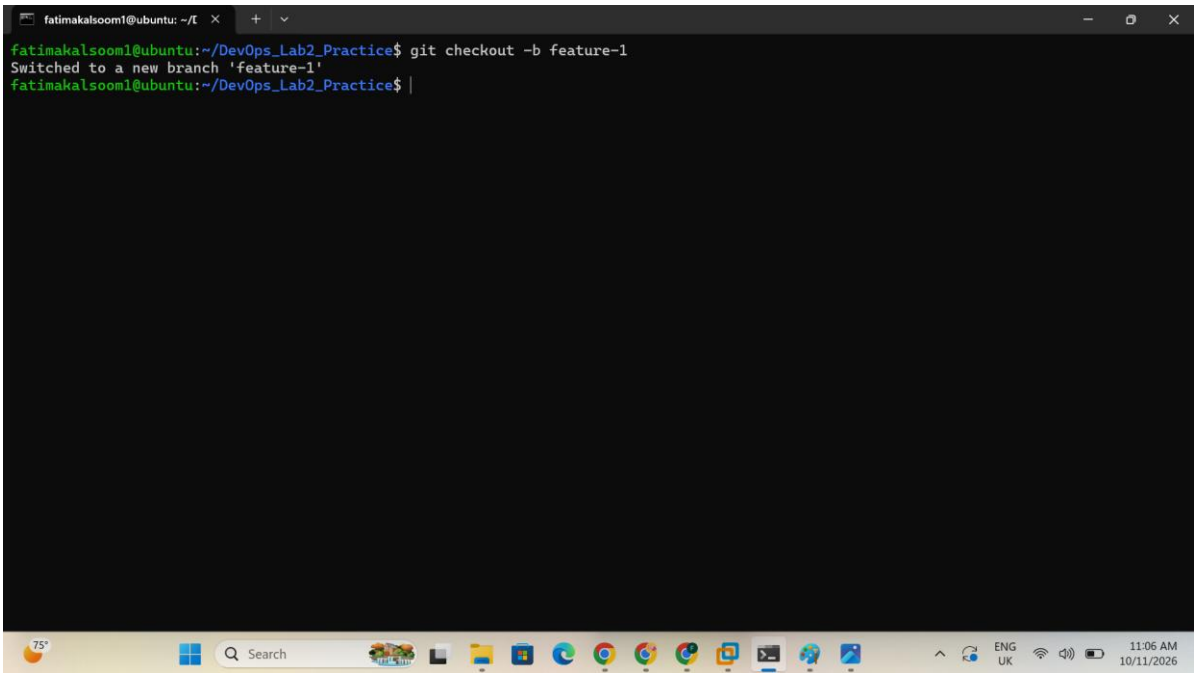
Branch pulled and visible in the local branch list.

Task 8

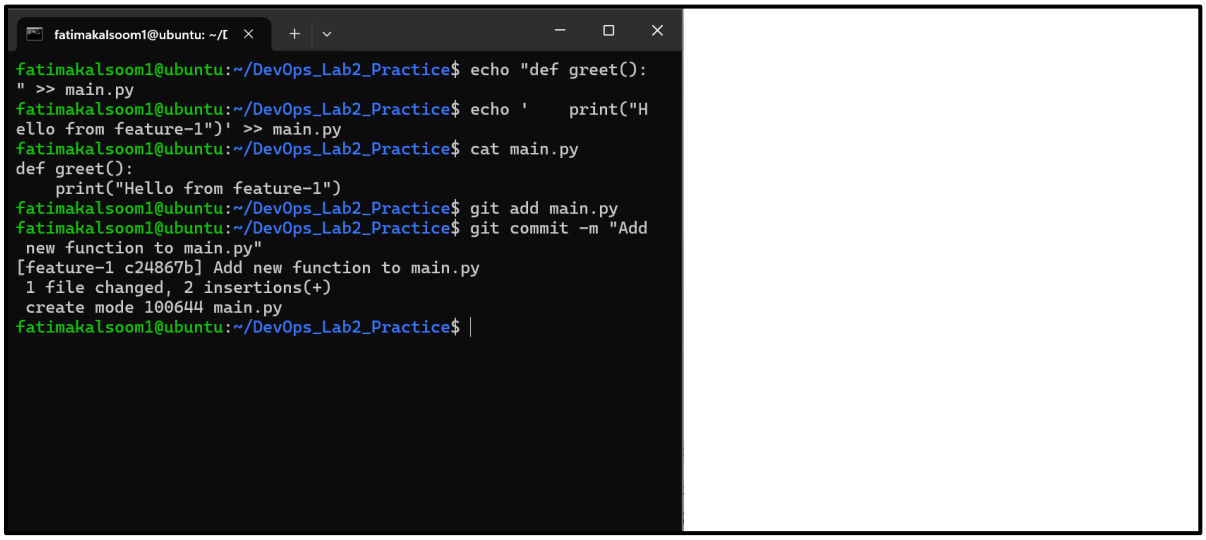


Branch feature/db-connection created and pushed to remote.

Task 9



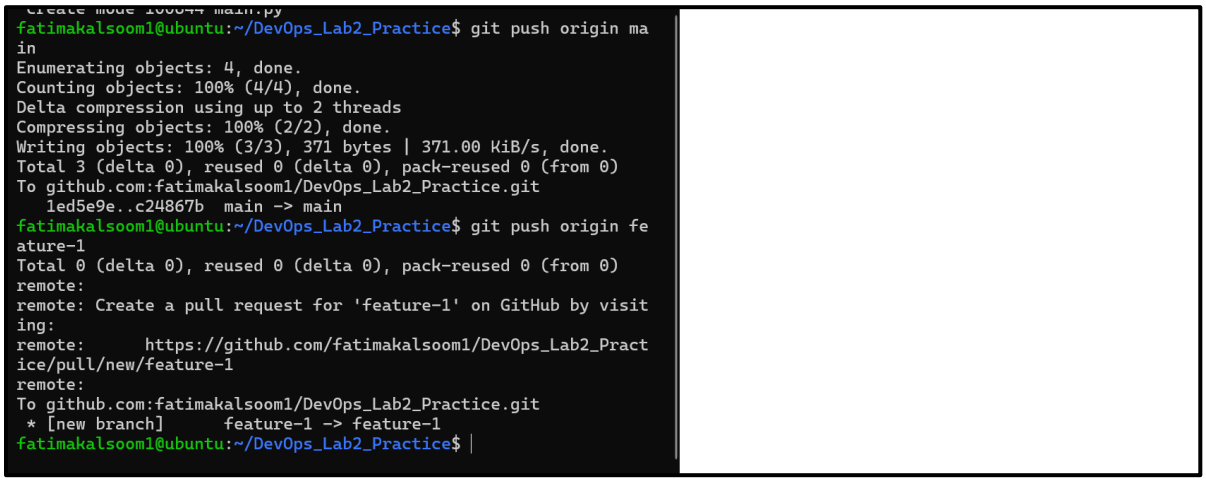
New branch feature-1 created and checked out.



New function added to main.py and committed on feature-1.

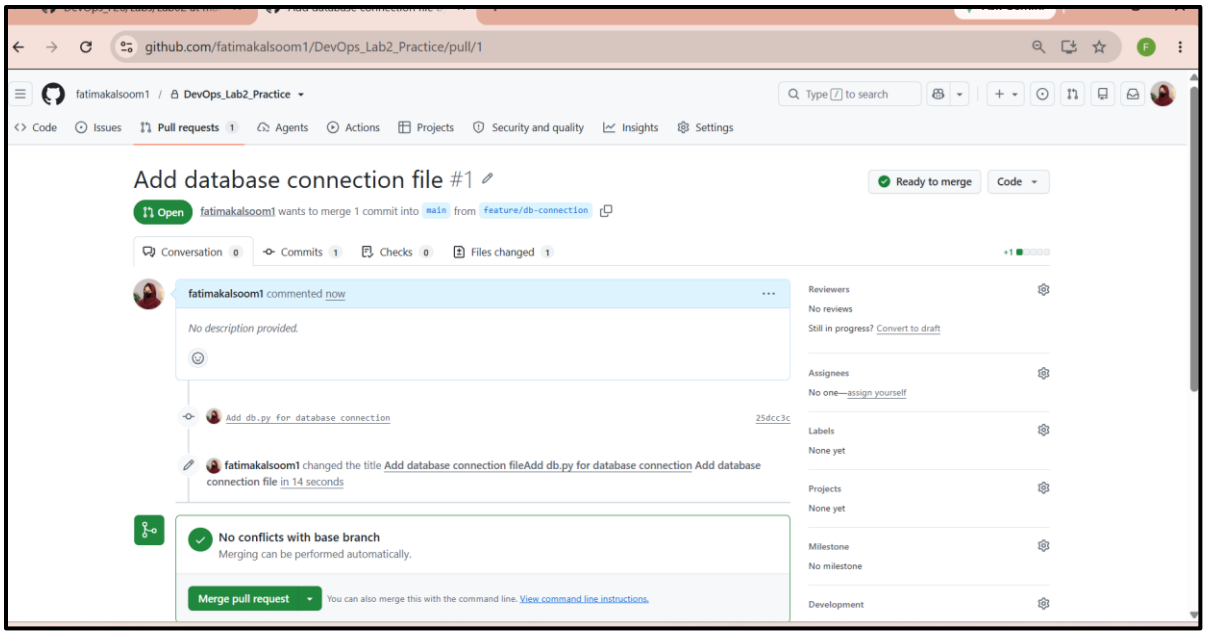


feature-1 merged into main.

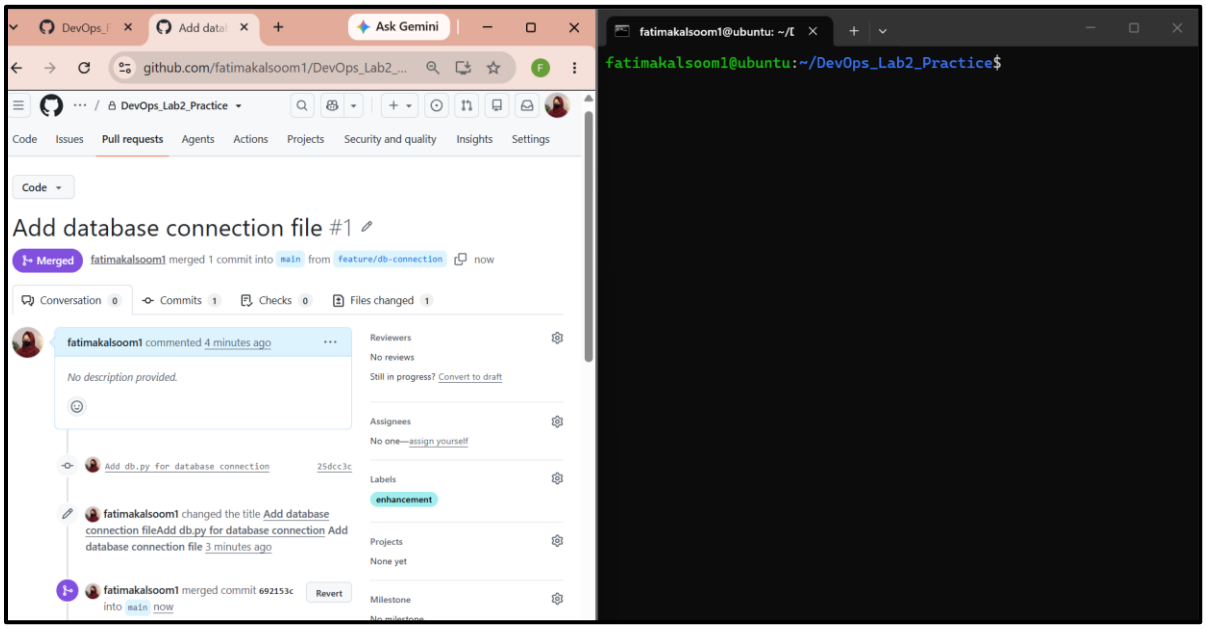


main and feature-1 pushed to GitHub.

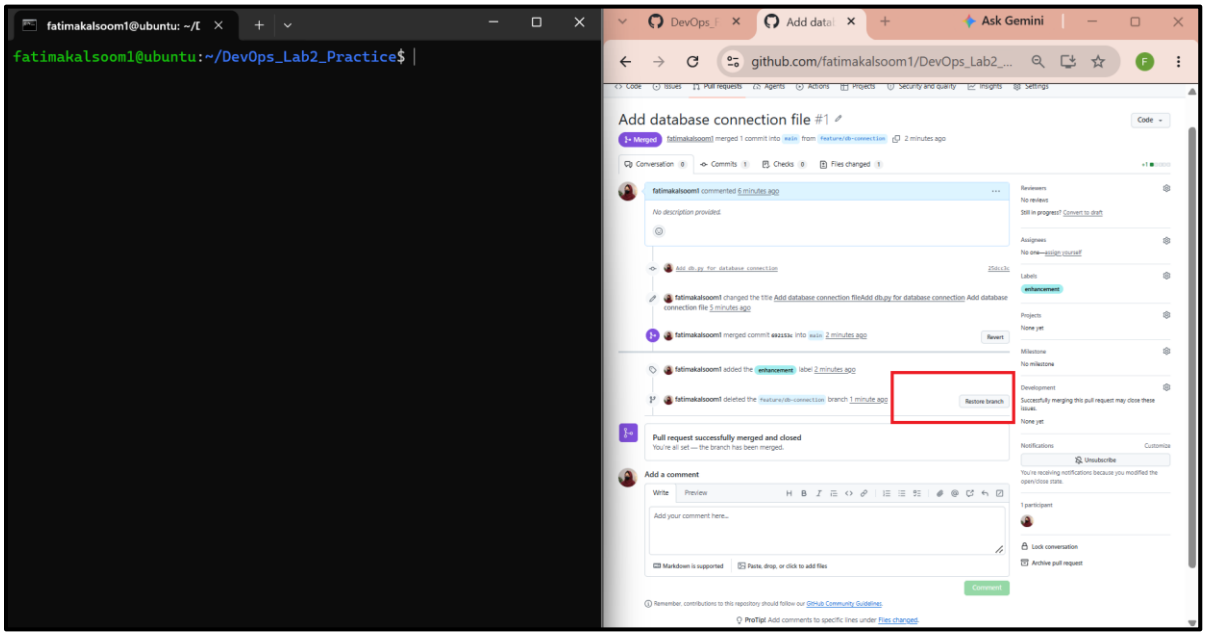
Task 10



Pull request created from feature/db-connection to main.

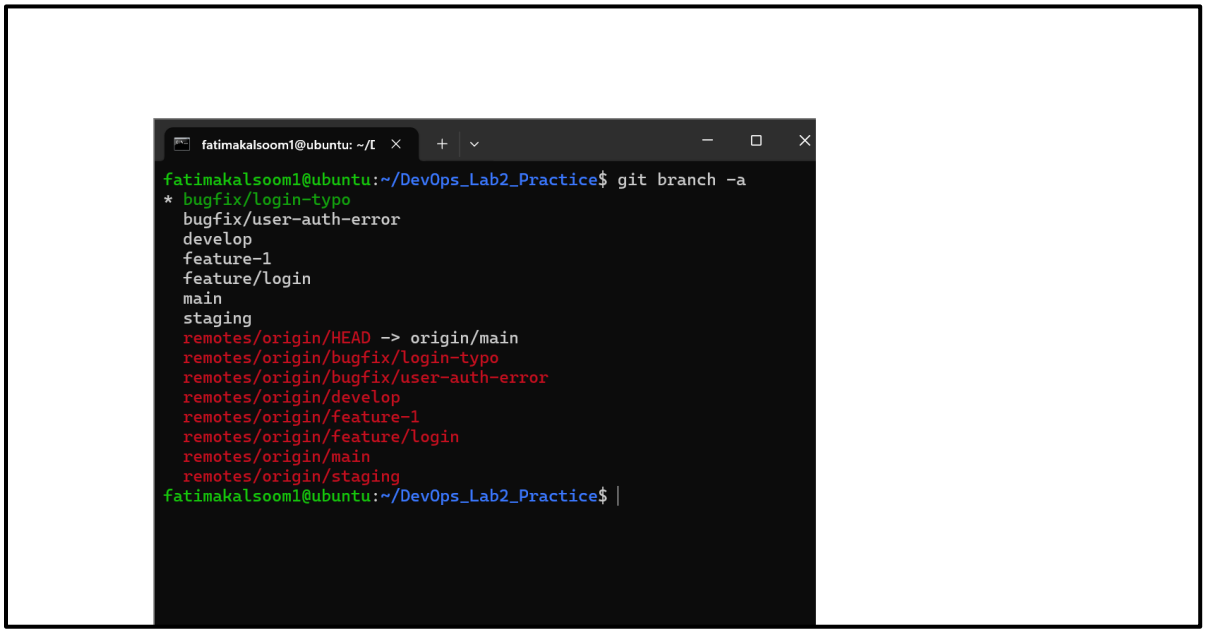


Pull request reviewed and merged through the GitHub interface.

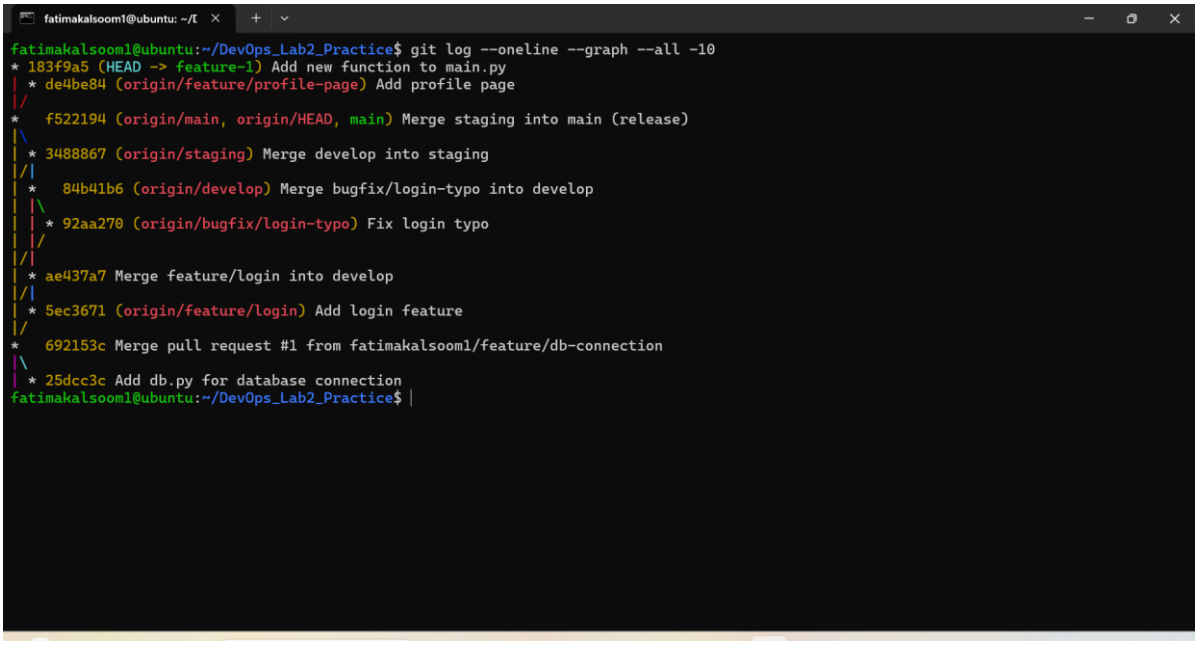


feature/db-connection deleted after the merge.

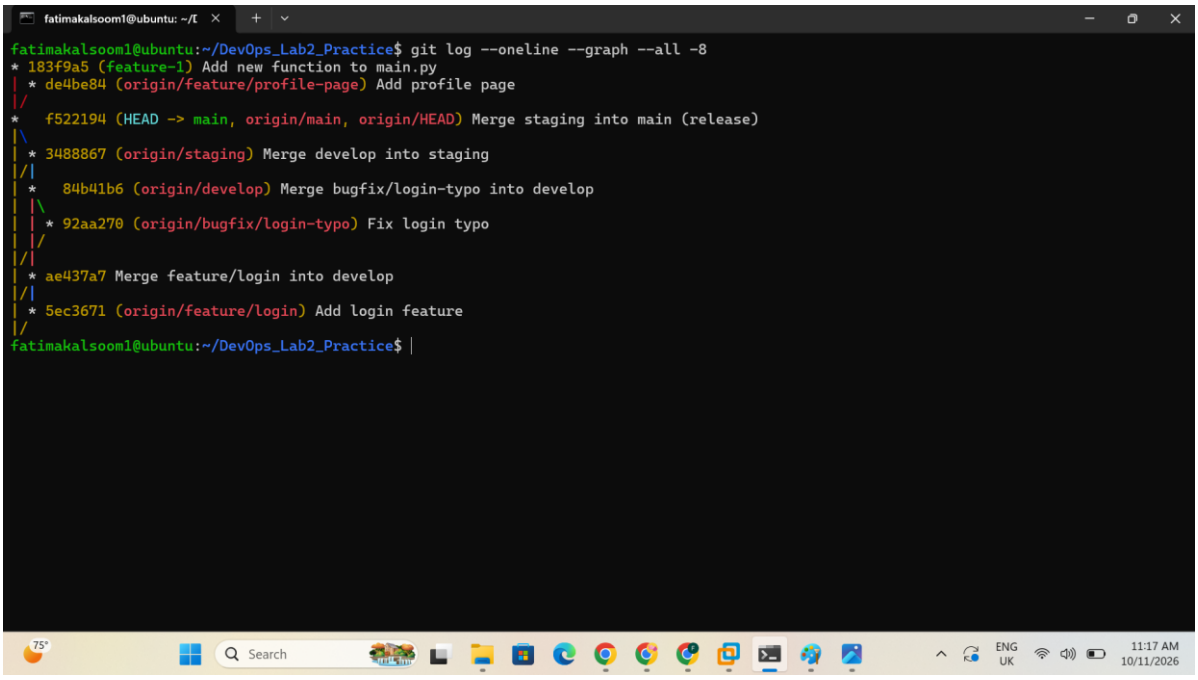
Task 11



Branch list showing main, develop, staging, feature and bugfix branches.

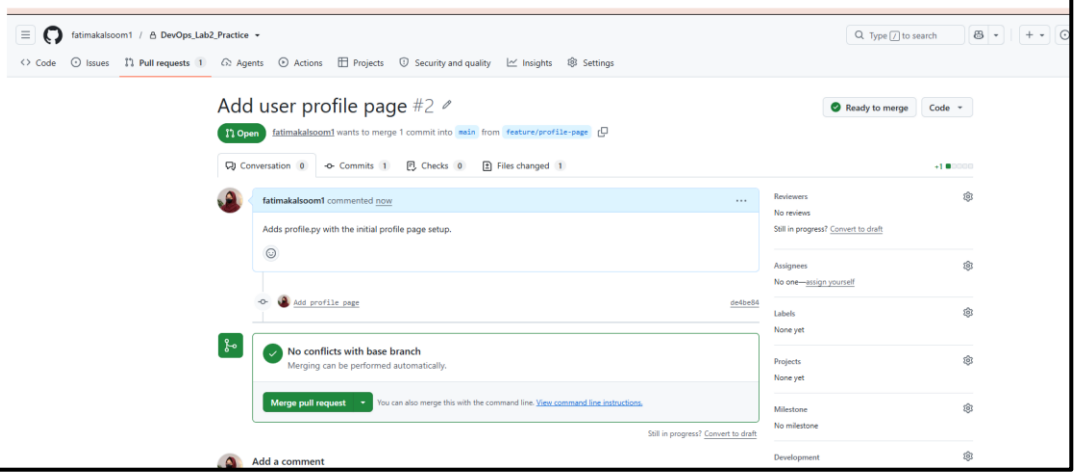


Feature and bugfix branches merged into develop, then develop merged into staging.

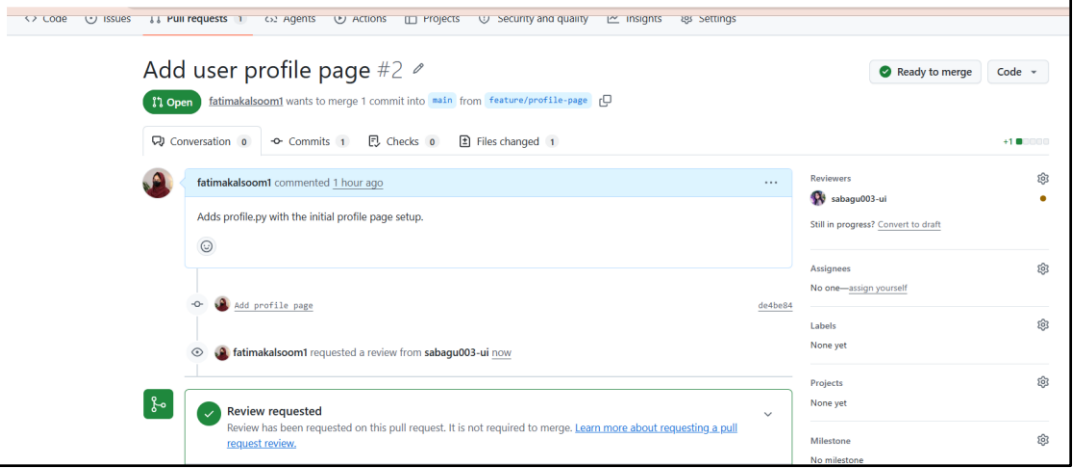


staging merged into main (production release).

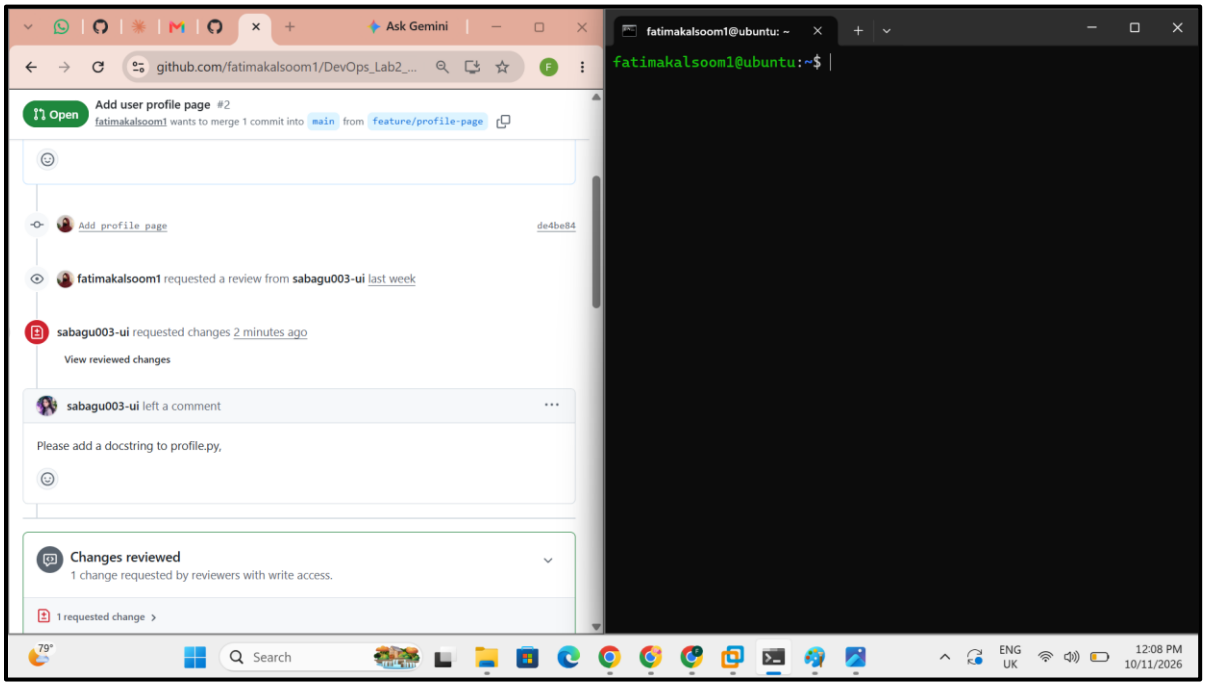
Task 12



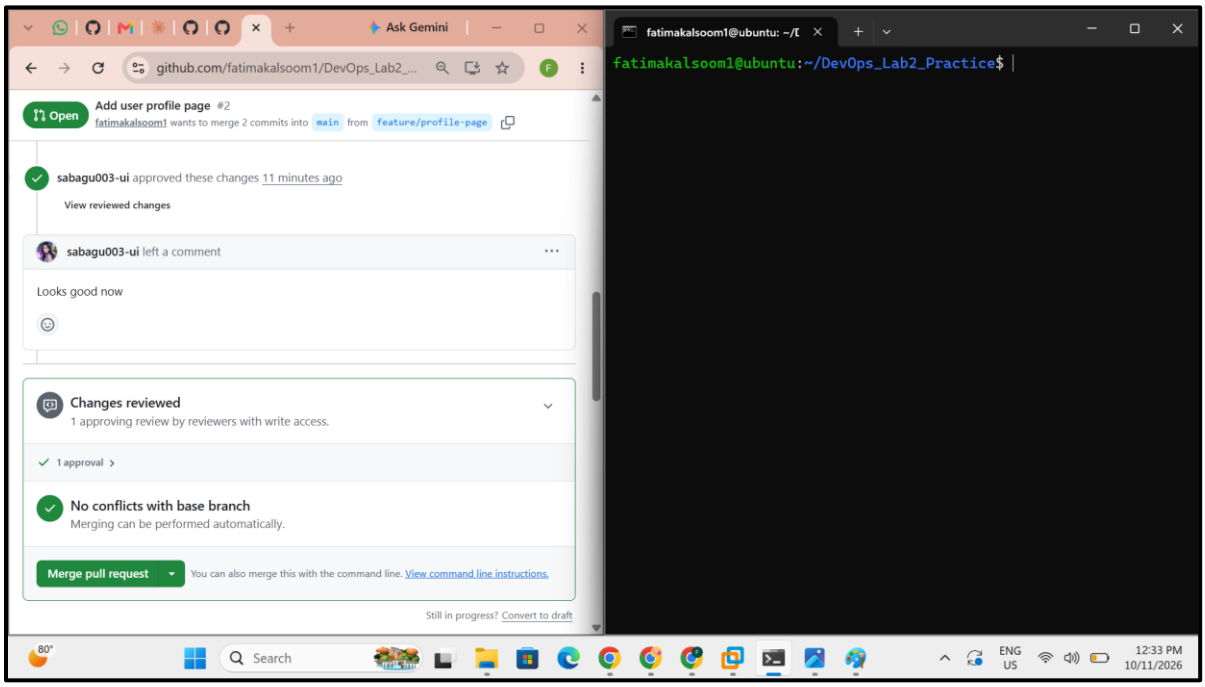
Pull request with a clear title and description.



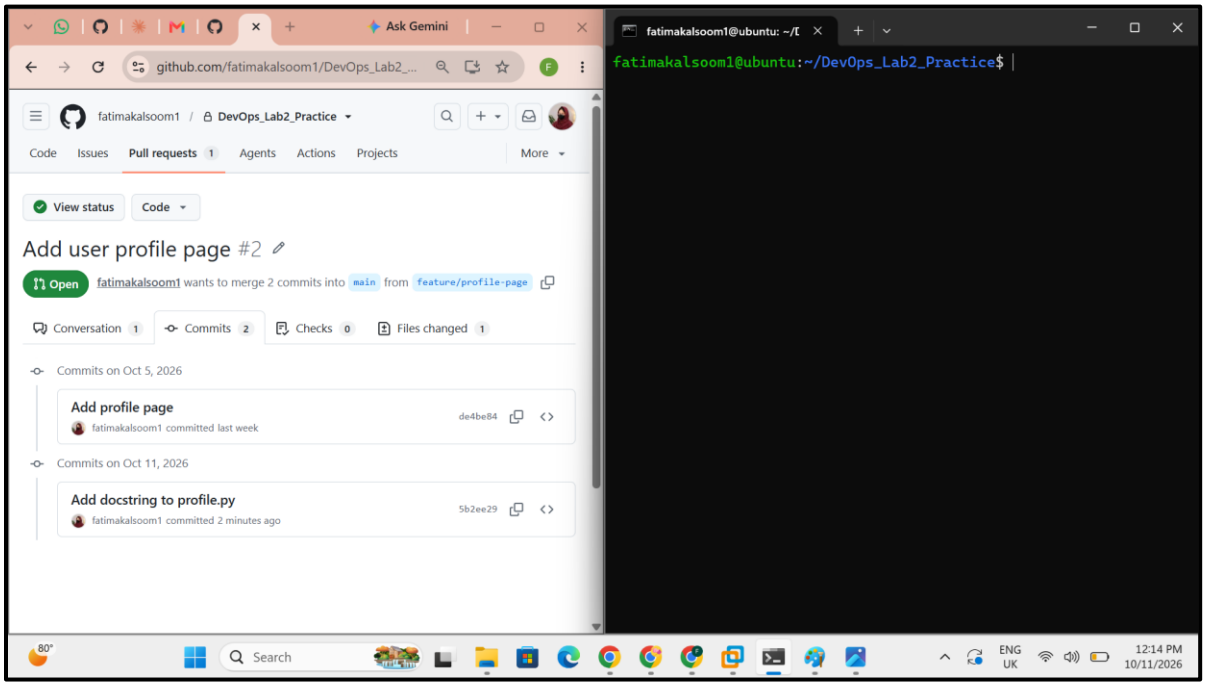
Reviewer assigned to the pull request.



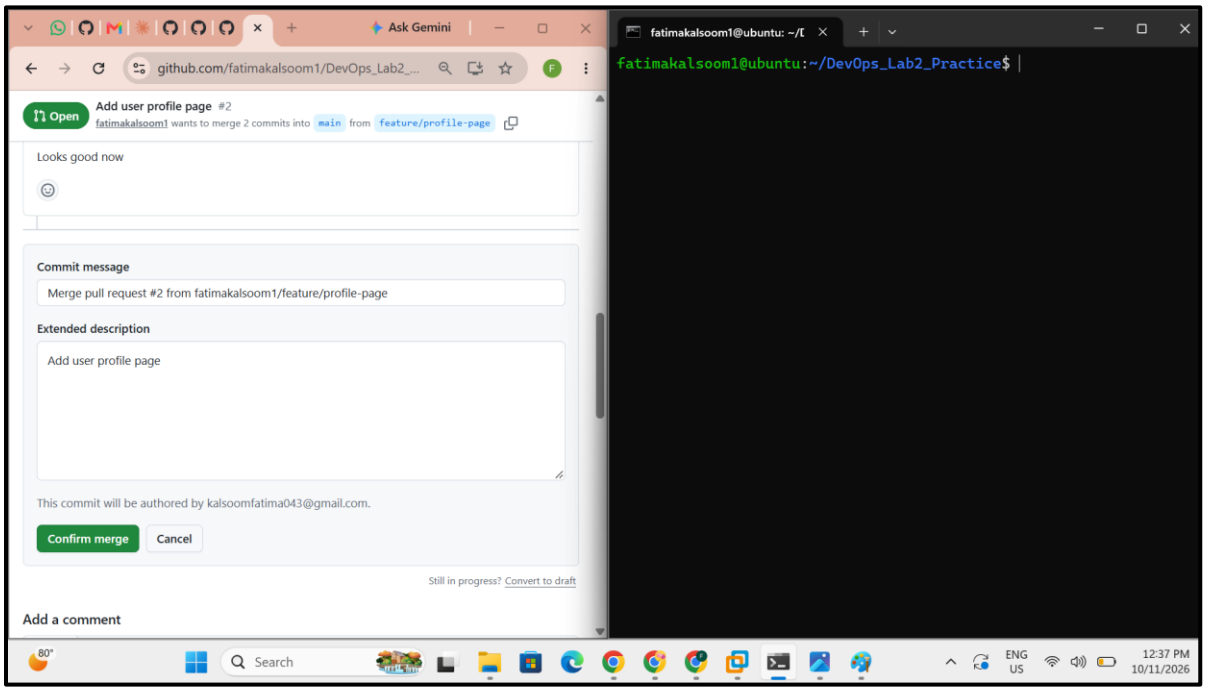
Reviewer requested changes with a comment.



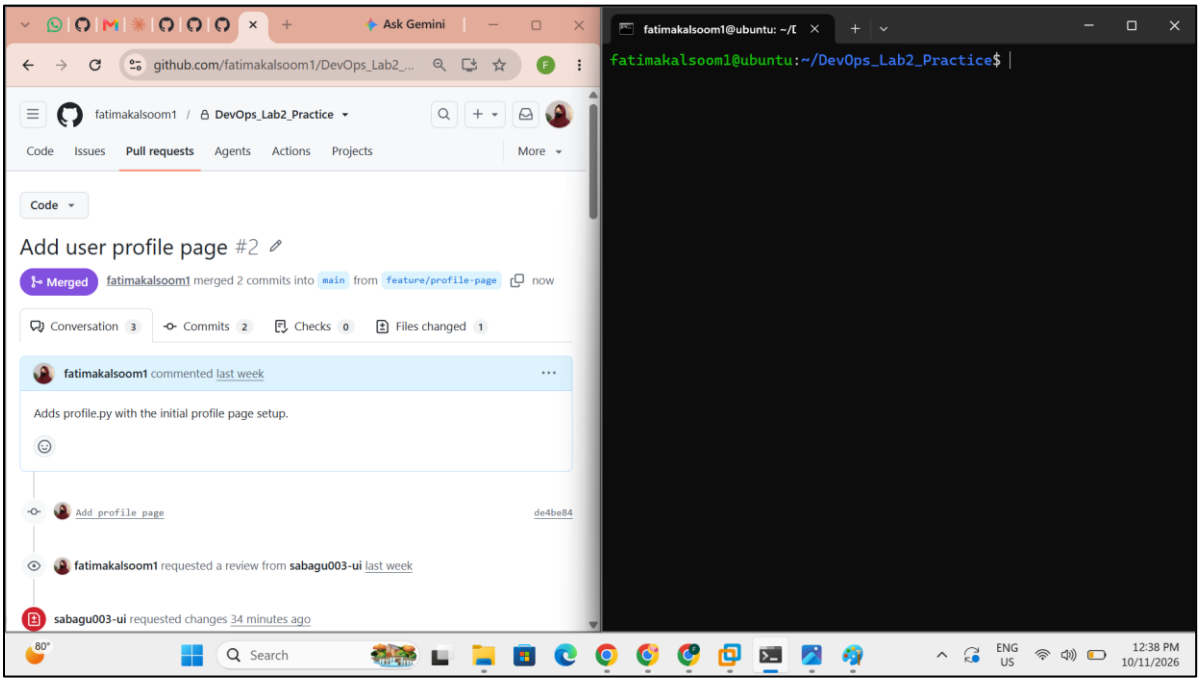
Practice pull request closed (rejected).



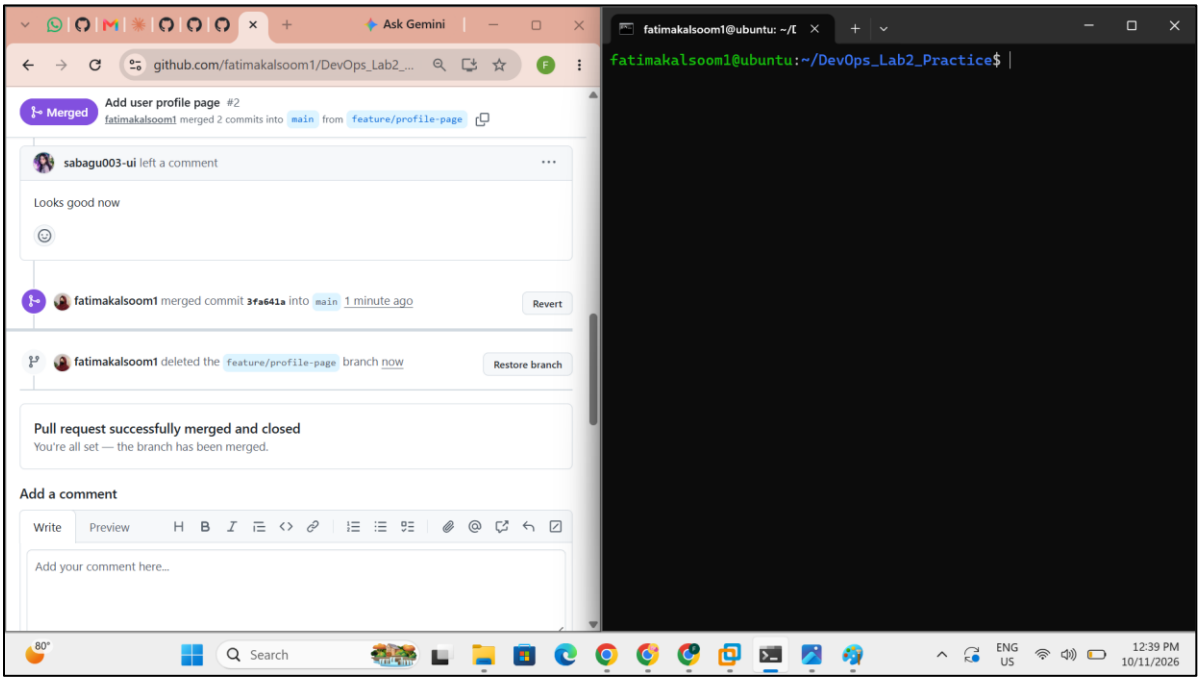
Fix pushed to the same branch; pull request updated with the new commit.



Merge confirmation screen after reviewer approval.

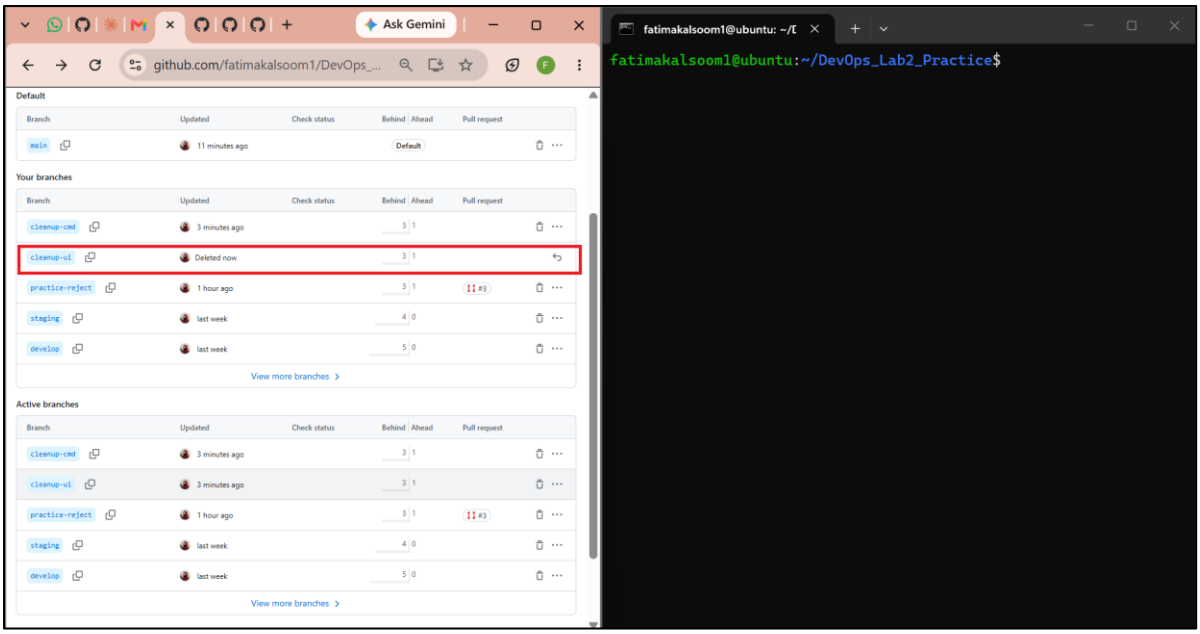


Pull request successfully merged.

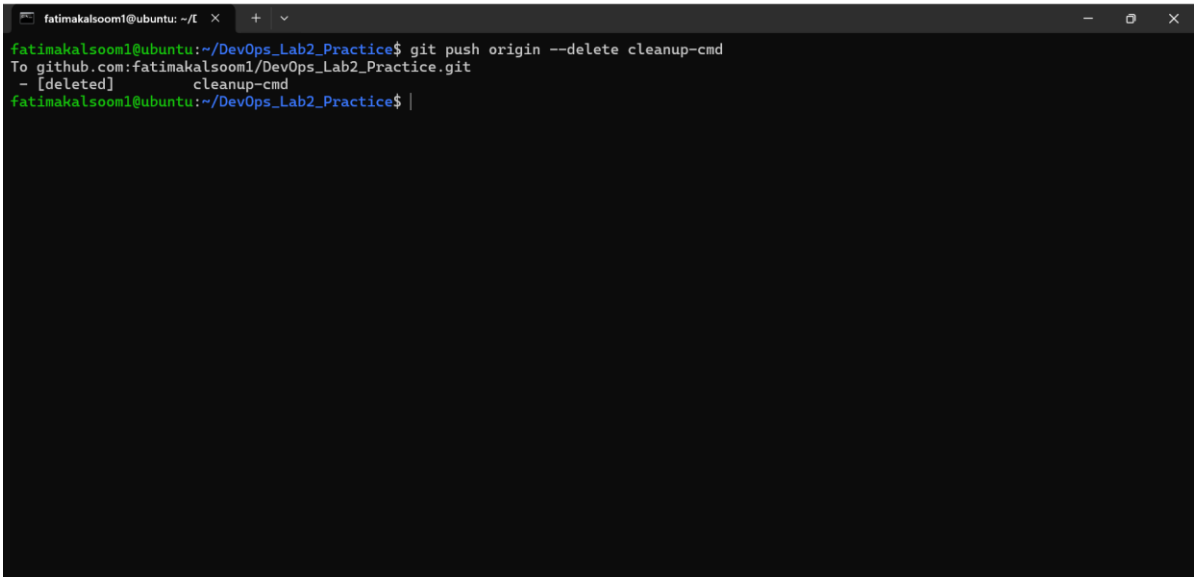


Merged pull request branch deleted.

Task 13

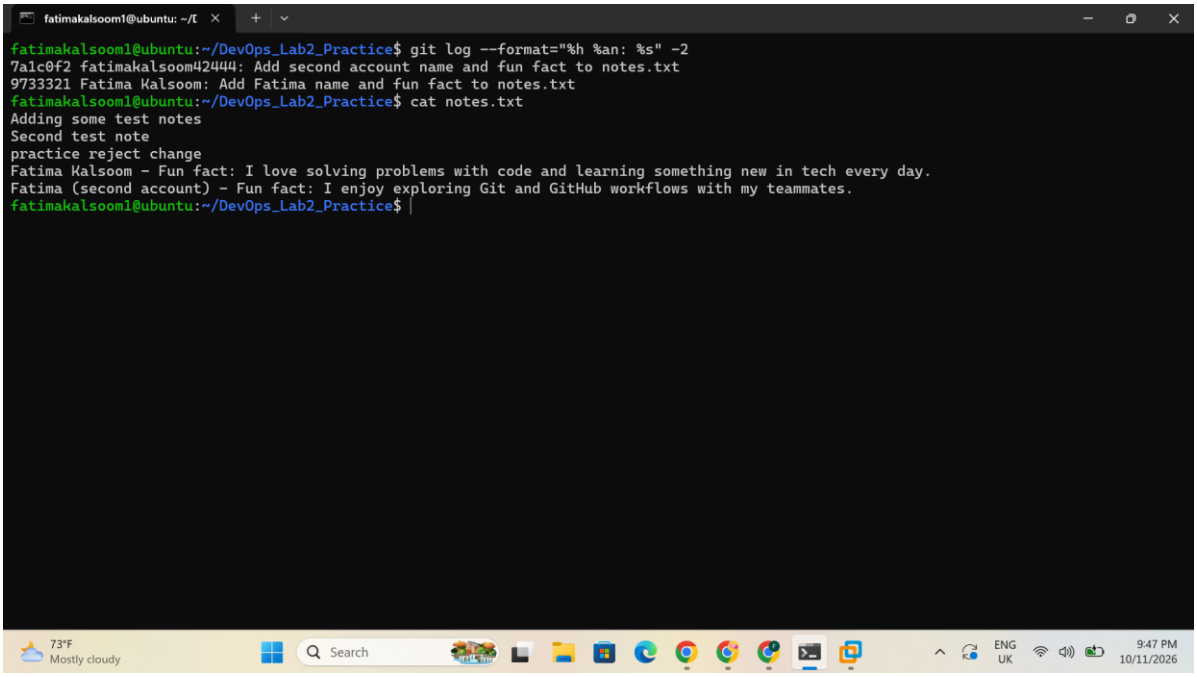


Remote branch deleted through the GitHub interface.

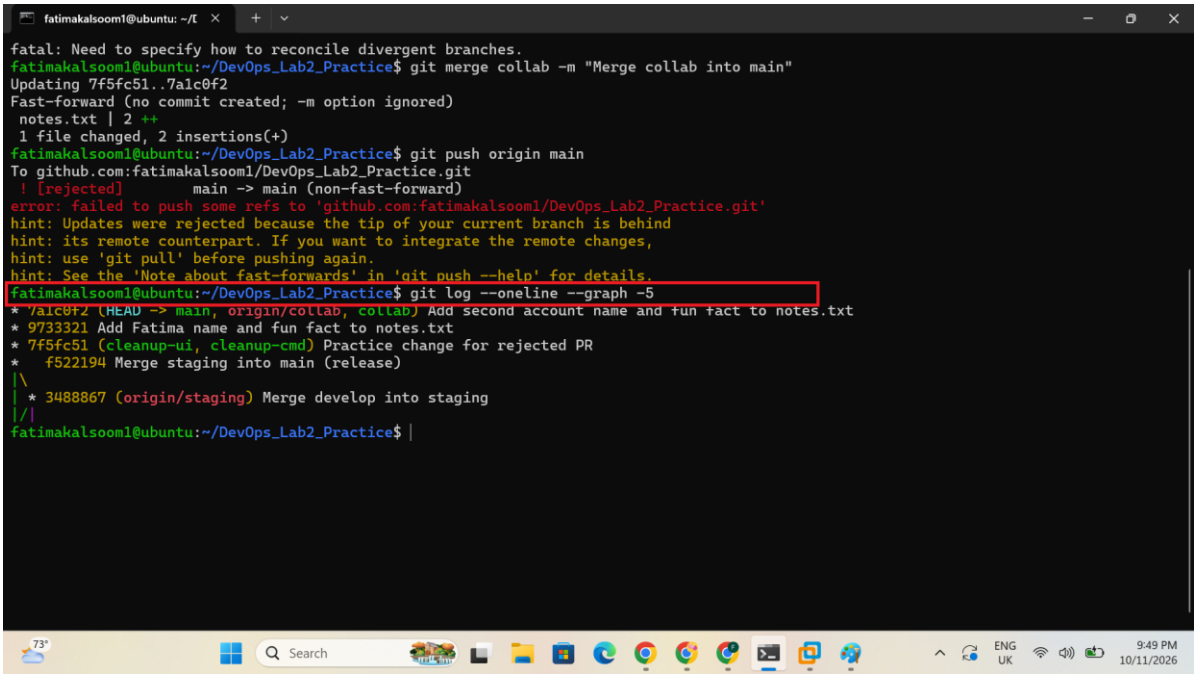


Remote branch deleted from the terminal.

Bonus Task

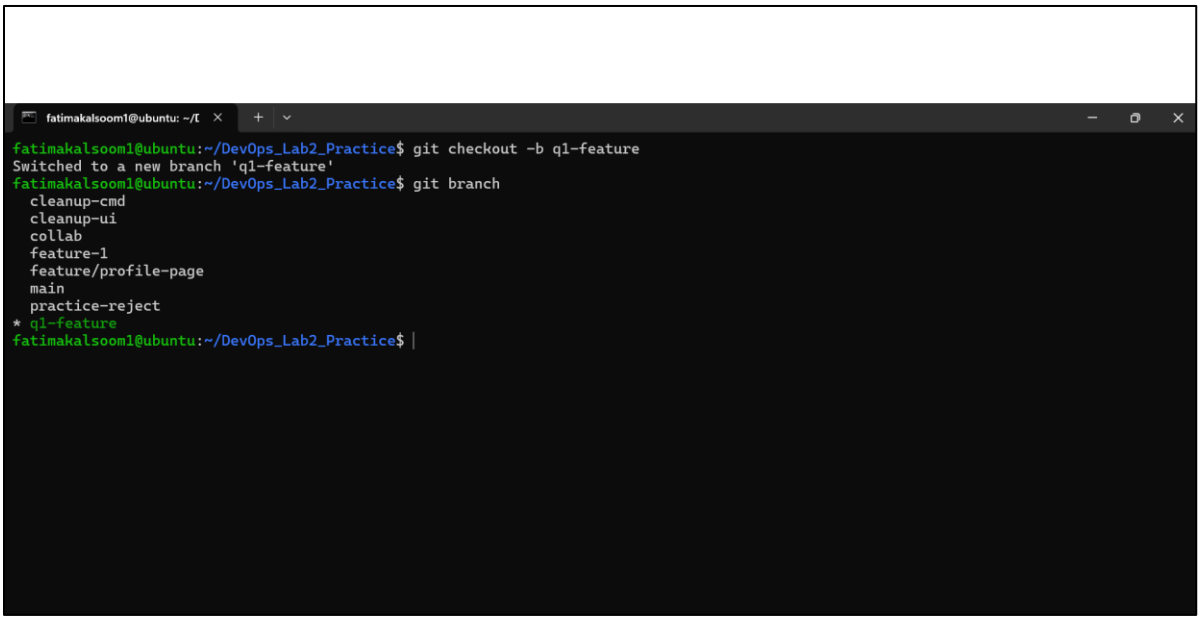


Two accounts committed their name and a fun fact to notes.txt on the collab branch.

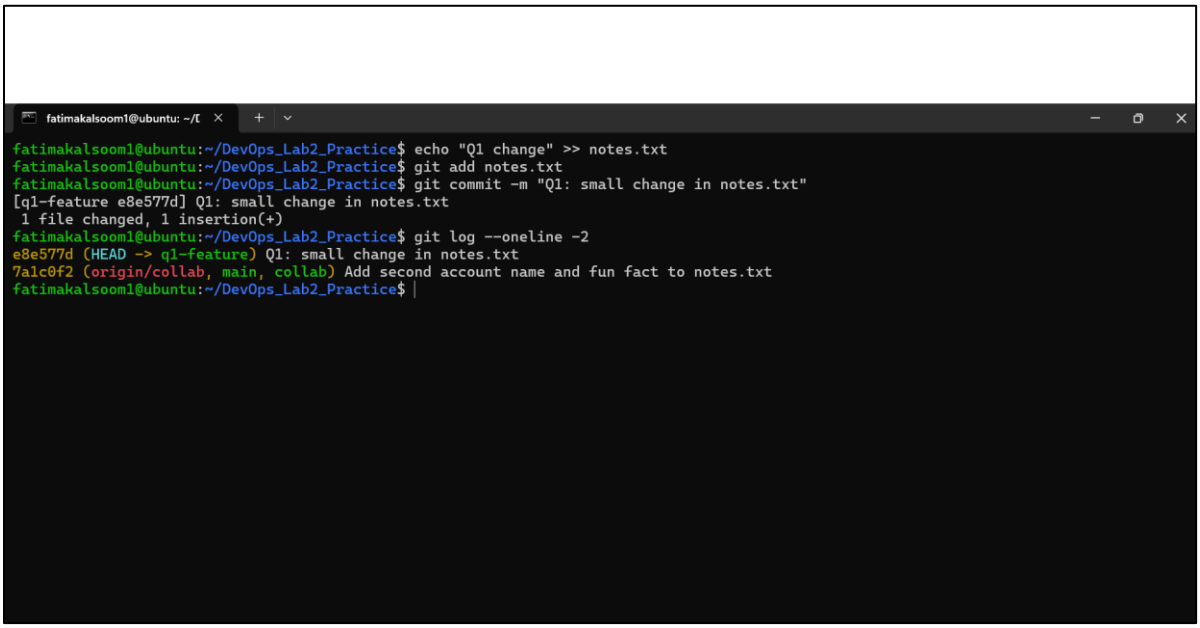


collab branch merged into main and pushed.

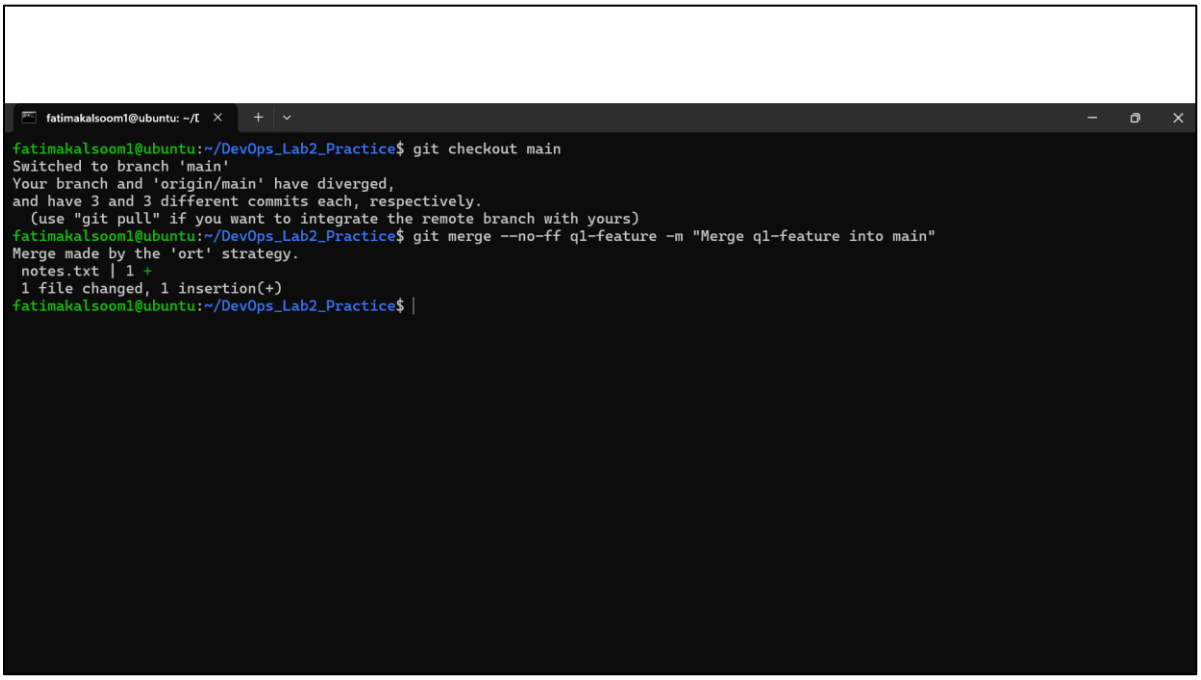
Exam Question 1



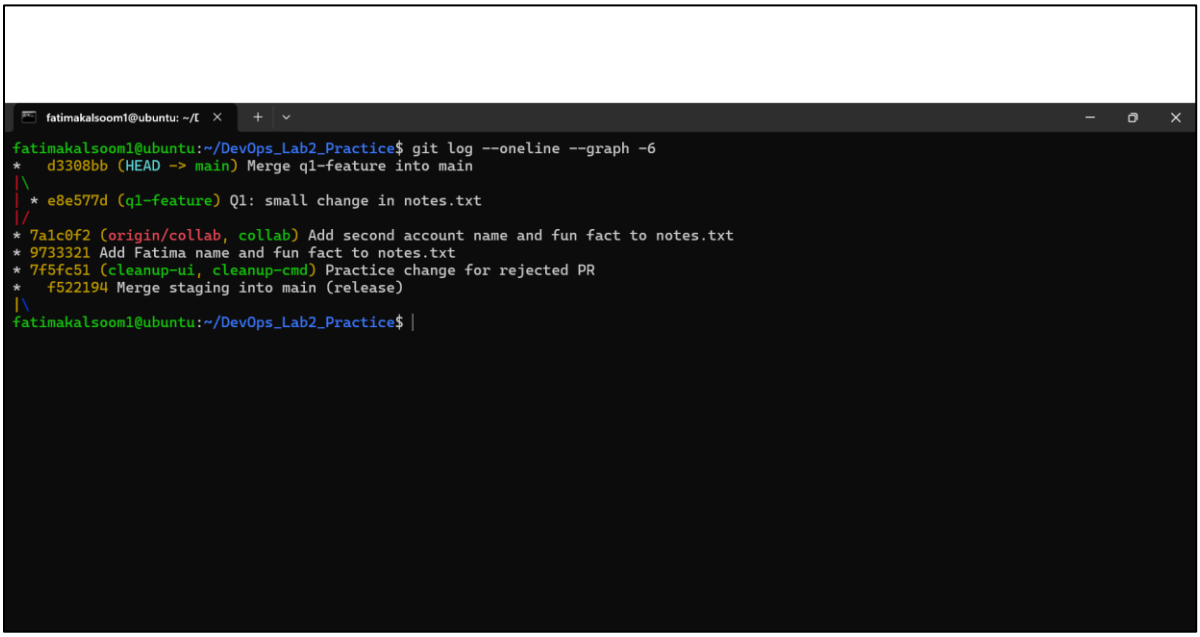
New branch created for the exam task.



Small change made and committed on the new branch.

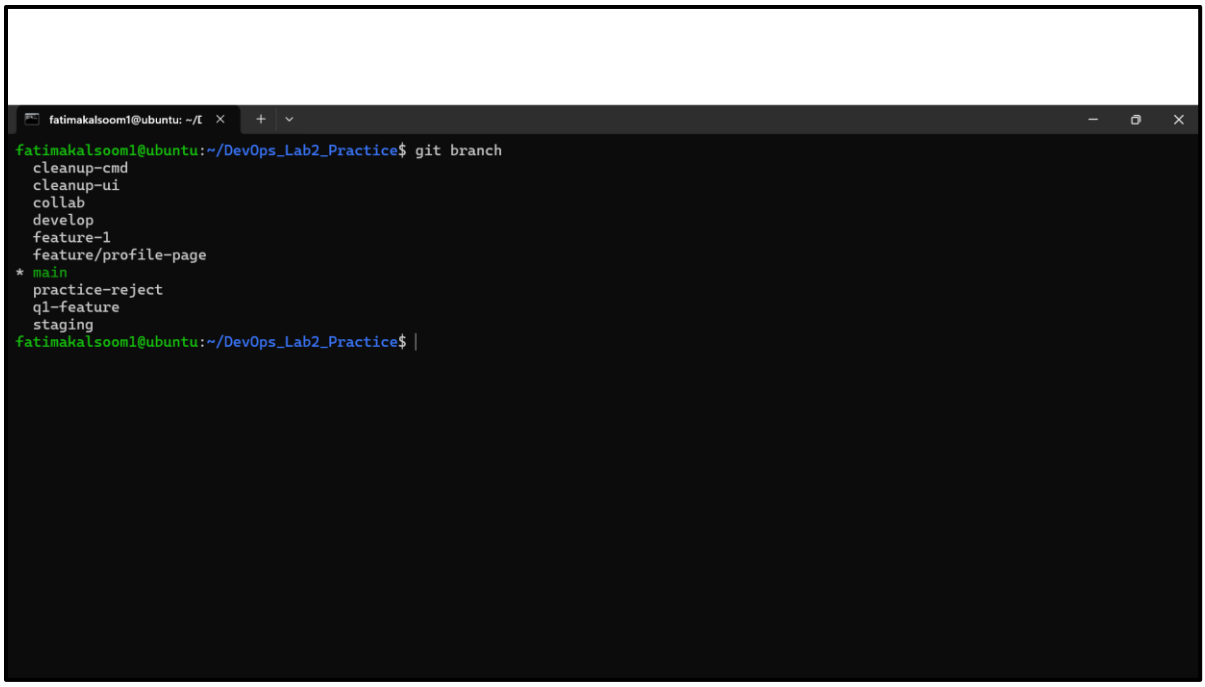


Branch merged into main.

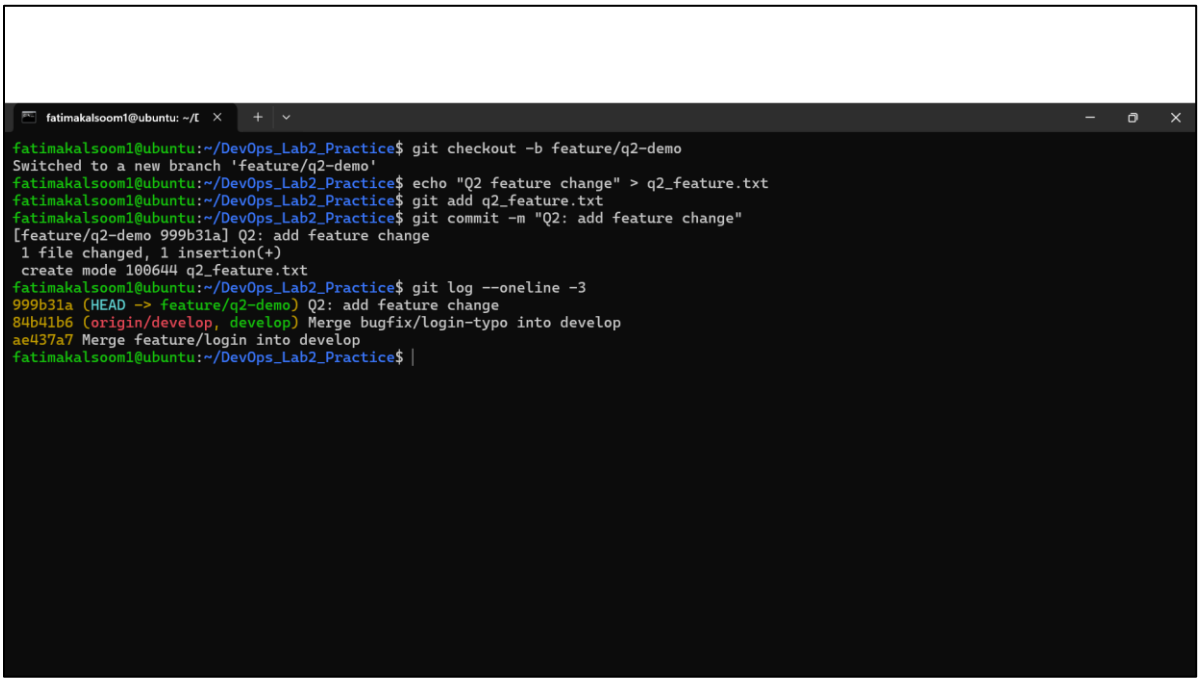


Commit history graph verifying the merge.

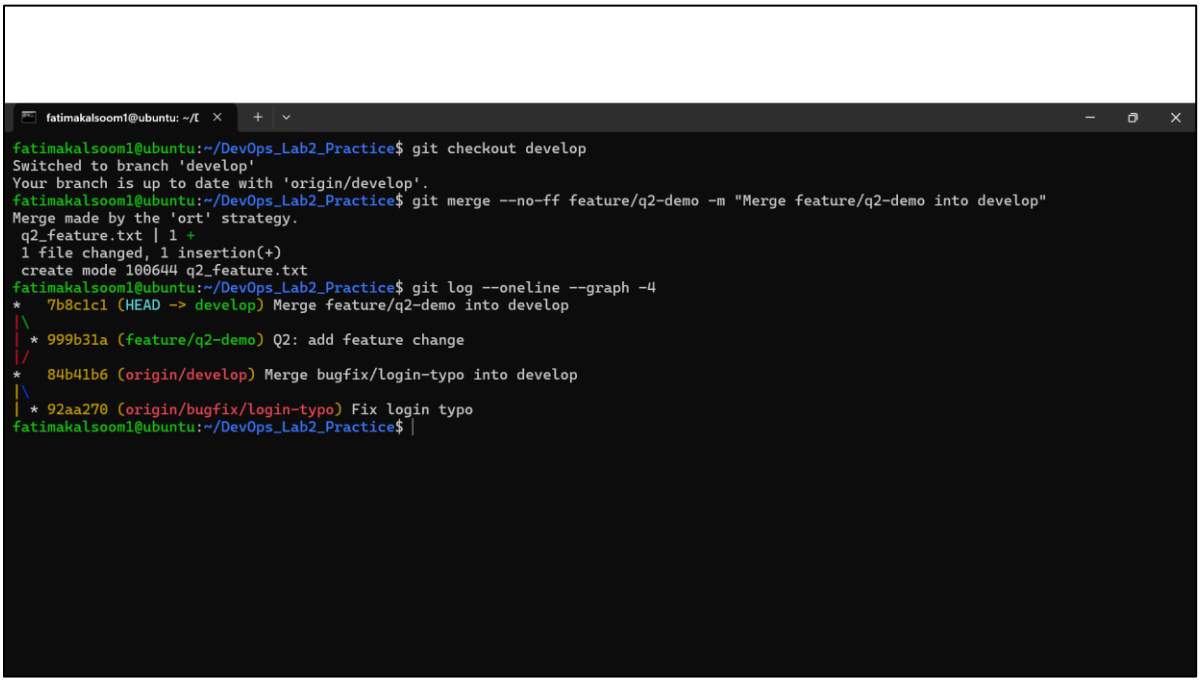
Exam Question 2



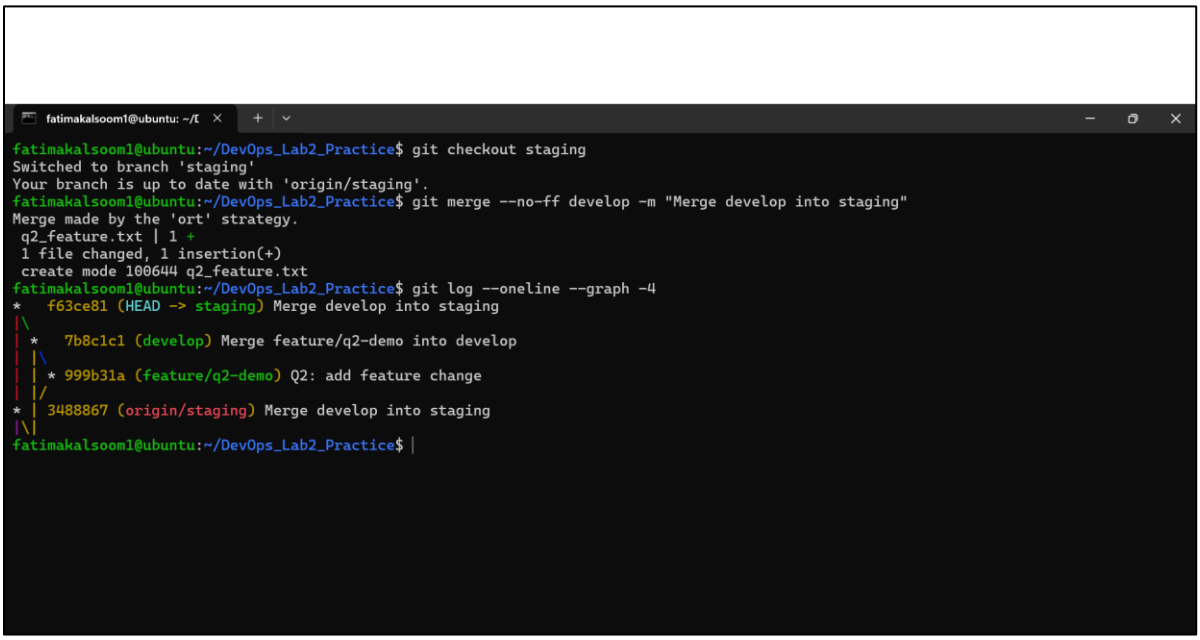
Branches main, develop and staging set up.



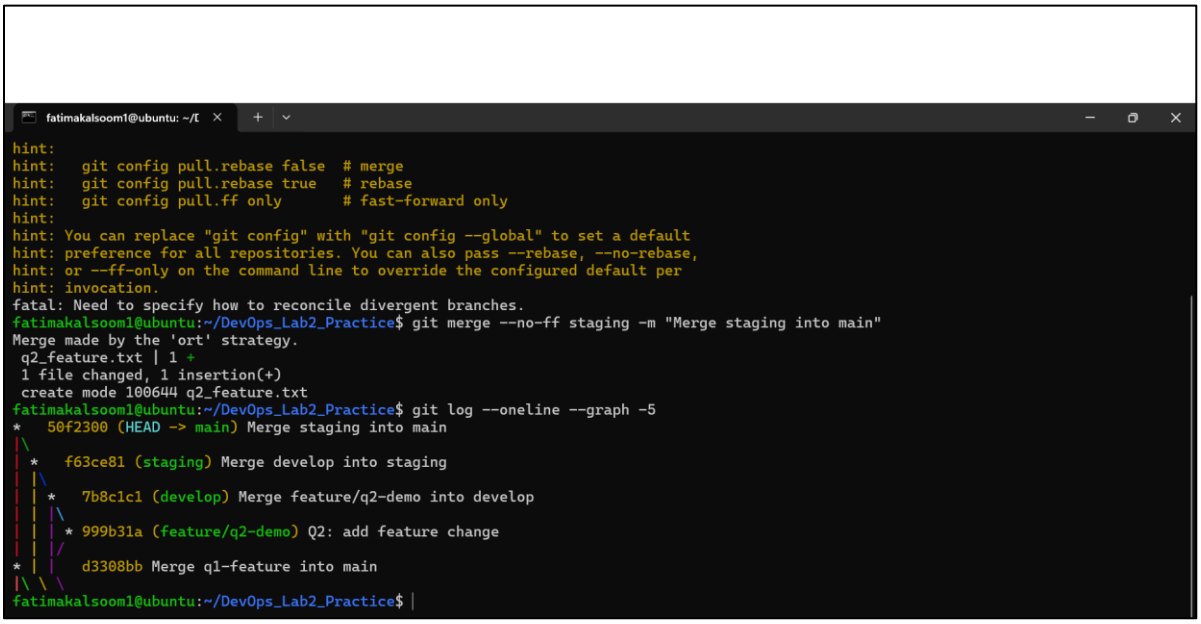
Feature branch created from develop with a commit.



Feature branch merged into develop.



develop merged into staging.

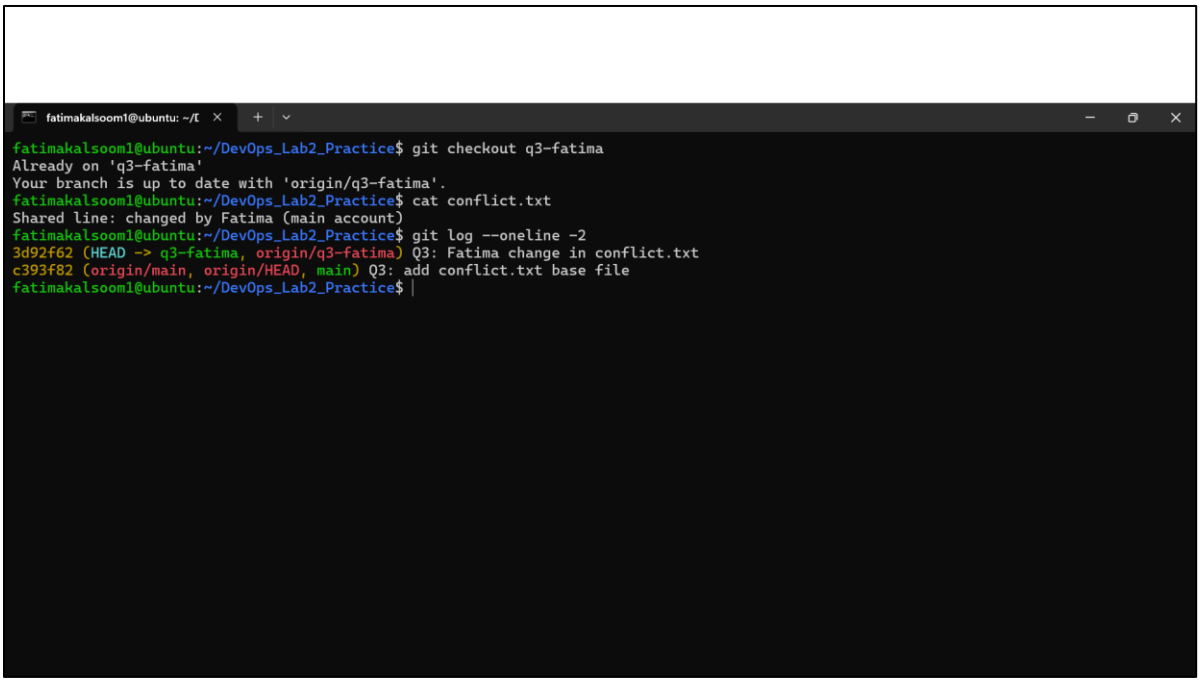


staging merged into main.

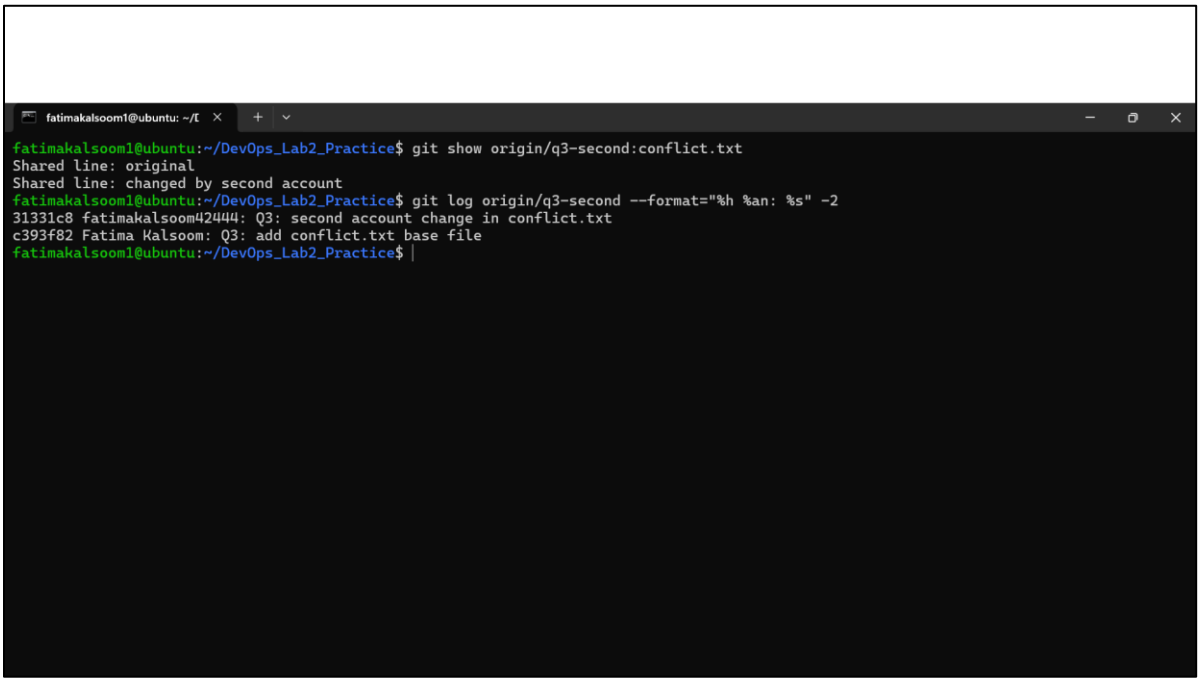


Updated file present in develop, staging and main.

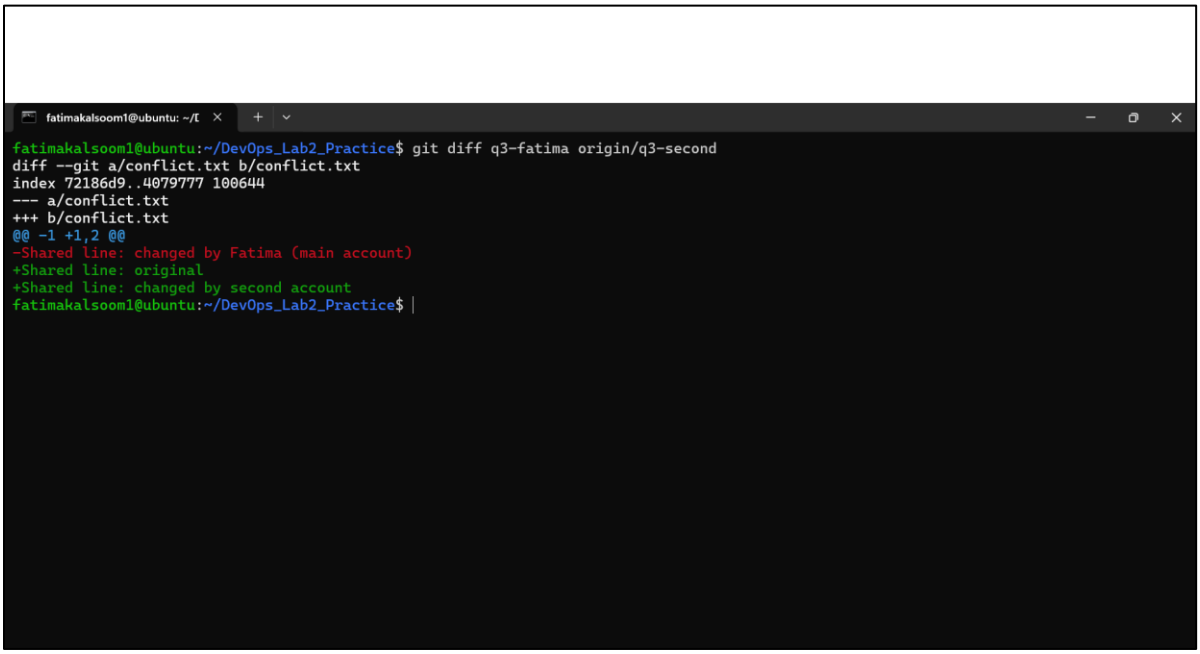
Exam Question 3



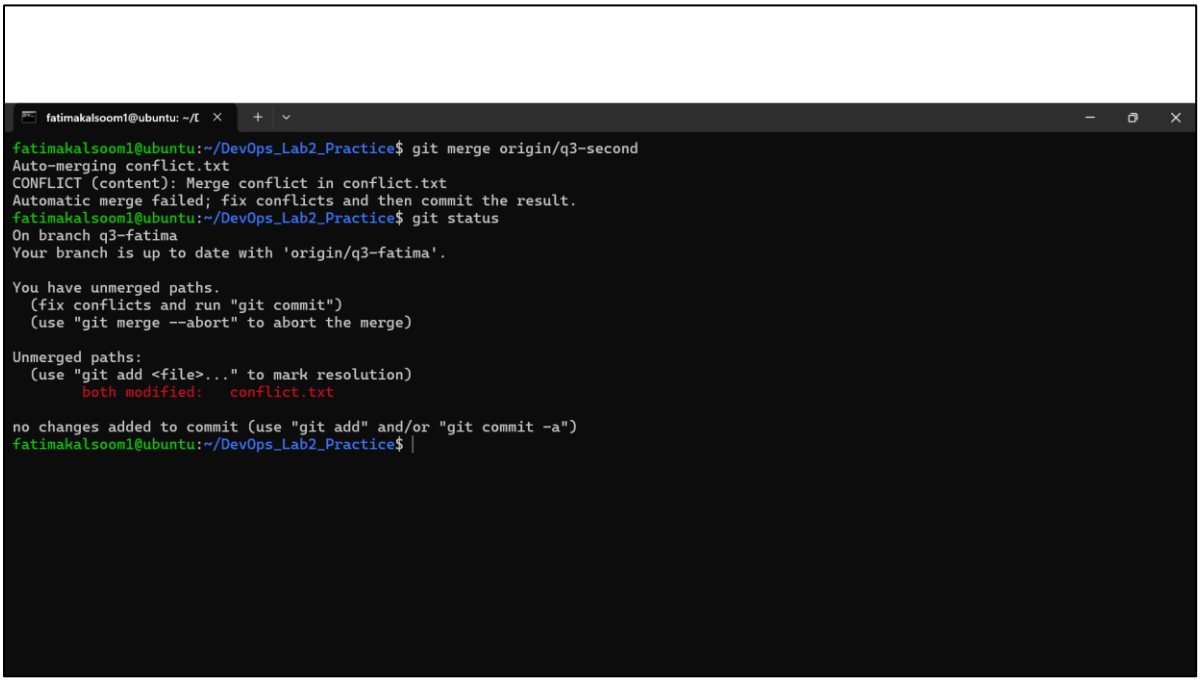
First contributor's change to conflict.txt on its branch.



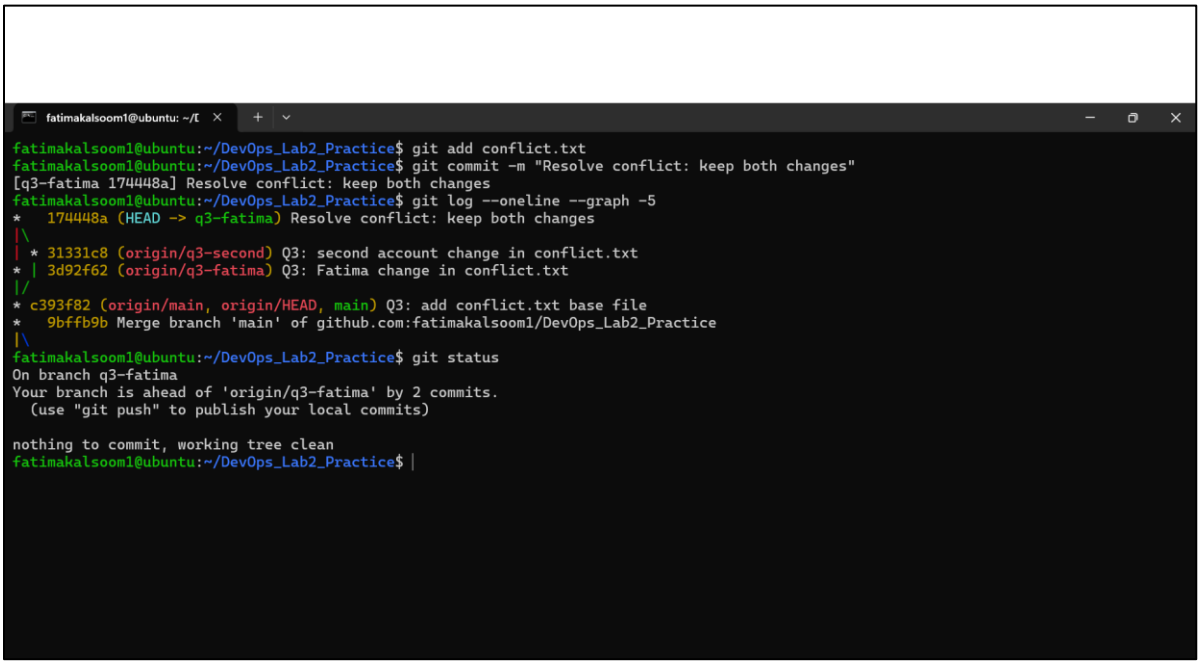
Second contributor's change to the same file on a separate branch.



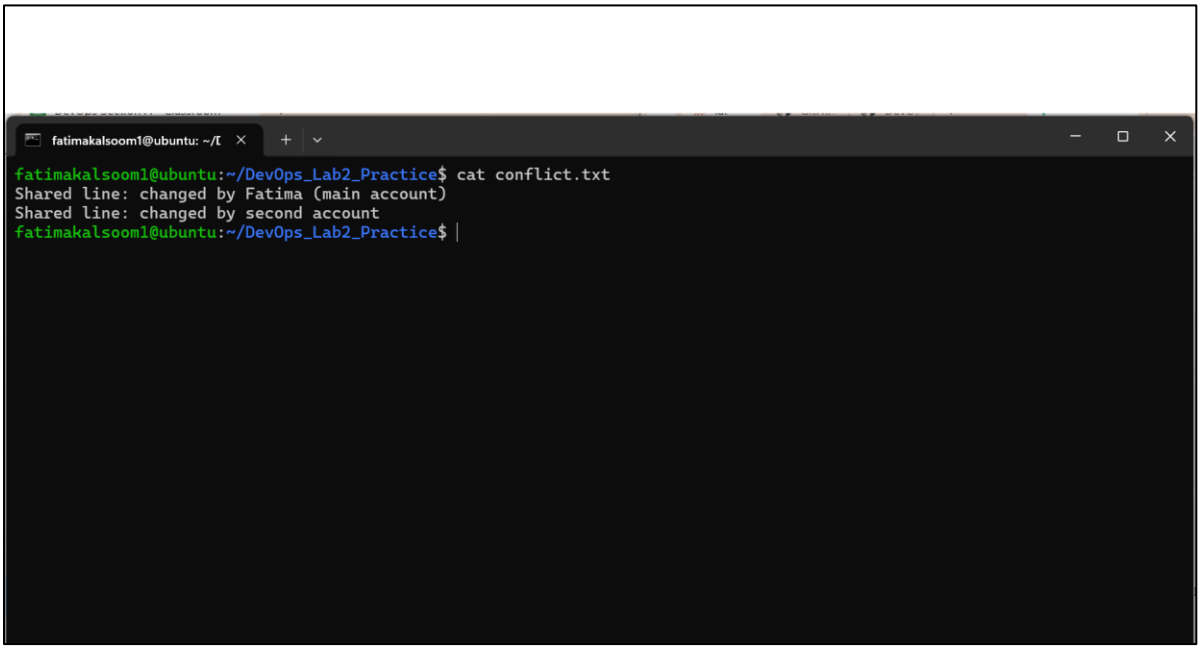
Difference between the two branches.



Merge conflict reported by Git.



Conflict resolved and merge committed.



Final conflict.txt containing both contributors' changes.